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Chapter 18

GELFAND-P000919

13.10.80 Gelfand Sem.

I. M. Gelfand. Consider the equation

$$f(z_1, \dots, z_n; w_1, \dots, w_n) = 0 \tag{1}$$

Example: consider a 2nd-order diff. eq.

$$y'' = f(x, y, y') \tag{2}$$



Original source details: GELFAND-P000919

Uncertain reading. A superscript mark above the second argument y is crossed out. Its reading is unresolved.

Then

$$y = \varphi(x, \xi, \eta) \tag{3}$$

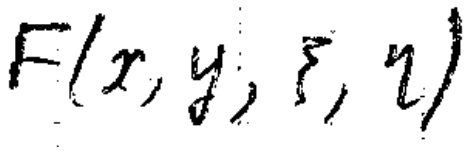


Original source details: GELFAND-P000919

Uncertain reading. The heavily traced letter reads φ.

Conversely, if we have family (3), a 2nd-order diff. eq. is constructed from it.
more generally:

$$F(x, y, \xi, \eta) = 0 \tag{4}$$



Original source details: GELFAND-P000919

Uncertain reading. In (4) there is a comma after y.

One can regard ξ, η as a point and construct the dual diff. eq.

(1) defines an n-param. family of hypersurfaces.

(5) Problem: find the geometry assoc. with this family.

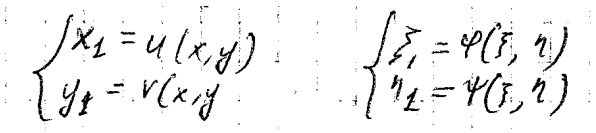
We can make changes of variables. E.g., in the eq. of straight lines

$$\xi x + \eta y = 1 \tag{6}$$

(7) Problem: how can we tell whether family (4) reduces to the family of straight lines (6).

Substitutions:

$$\begin{cases} x_1 = u(x,y) \\ y_1 = v(x,y \end{cases} \qquad \begin{cases} \xi_1 = \varphi(\xi,\eta) \\ \eta_1 = \psi(\xi,\eta) \end{cases}$$



Uncertain reading. The second line of the left-hand system has no closing parenthesis after y.

Original source details: GELFAND-P000919

(8) General definition: double fibration

(9) an element — a point and a curve passing through it

(curves from a 2-param. family)

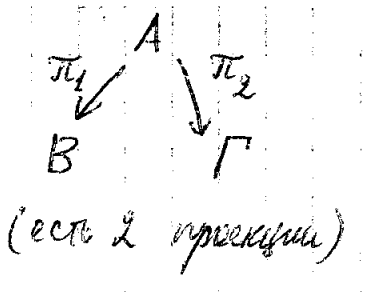
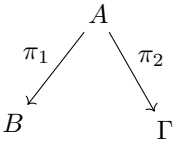


Diagram and typed description. Two projections labelled π_1 and π_2.



(there are 2 projections)

Original source details: GELFAND-P000919

(see the article by Gelfand, Gindikin, Shapiro in FA)

(10) Requirement: if

$$\begin{cases} & \pi_1 x = \pi_1 x' \\ \text{and} & \pi_2 x = \pi_2 x' \end{cases}, \quad \text{then } x = x'$$

a stronger requirement:

$$(11) \qquad A \subset B \times \Gamma \quad \text{as a submanifold}$$

$$x \mapsto (\pi_1 x, \pi_2 x)$$

Problem: classify up to diffeomorphism

$$f(z, w) = 0, \tag{12}$$

up to changes

$$\varphi(z) = z_1, \qquad \psi(w) = w_1$$

(everything analytic).

(13) Another example: $A = G$ — a group, K, H — 2 subgroups,

$$B = H\backslash G, \qquad \Gamma = K\backslash G.$$

(14) Important example:

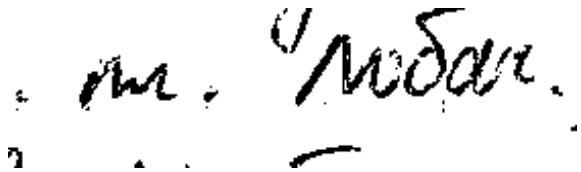
G — the group $GL(n, \mathbb{C})$,

K — subgr. of unitary matrices

H — —//— of triang. matrices

(15) Useful special case: $n = 2$.

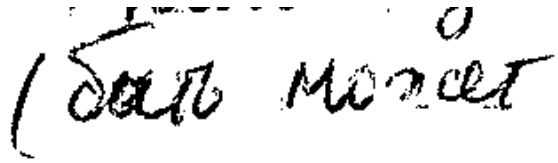
Then Lobach. geom. and horospheres arise.



Original source details: GELFAND-P000920

Uncertain reading. The abbreviation before ‘Lobach.’ reads «пл.», possibly ‘plane’.

(perhaps take 1 on the main diagonal in H)

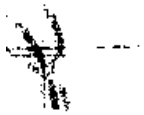


Original source details: GELFAND-P000920

Uncertain reading. The parentheses begin ‘perhaps’.

Work in Inventiones on Segre families.

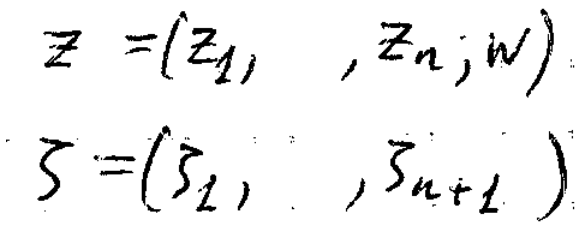
$$\tau(z,\zeta)=0 \tag{16}$$



Uncertain reading. Separate illegible strokes in the right margin level with (16).

Original source details: GELFAND-P000920

$$z=(z_1,\quad ,z_n;w)$$
$$\zeta=(\zeta_1,\quad ,\zeta_{n+1})$$



Uncertain reading. In both tuples, the spaces between the commas contain no visible ellipsis.

Original source details: GELFAND-P000920

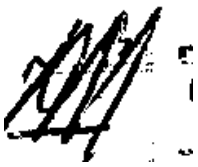
$$\theta=d_z\tau=\tau_\alpha\,dz^\alpha+\tau_w\,dw$$
$$\theta^\alpha=dz^\alpha$$
$$\theta_\alpha=d\zeta_\alpha$$

Under the map

$$(z,\zeta)\to(f(z),\bar f(\zeta)) \tag{17}$$

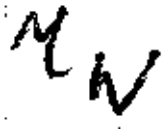
the forms transform as follows:

$$\theta^*=\theta,$$
$$\theta^{\alpha *}=\left(\frac{\partial f^\alpha}{\partial z^\beta}-\frac{\tau_\beta}{\tau_w}\frac{\partial f^\alpha}{\partial w}\right)\theta^\beta+\frac{1}{\tau_w}\frac{\partial f^\alpha}{\partial w}\theta, \tag{18}$$



Undeciphered passage. An illegible expression is crossed out after the minus sign and before the fraction with τ_β.

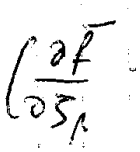
Original source details: GELFAND-P000920



Uncertain reading. In the last term the glyph before the subscript w resembles τ with an angular top. The reading τ_w is tentative.

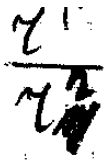
Original source details: GELFAND-P000920

$$\theta^*_\alpha = \left(\frac{\partial \bar{f}}{\partial \zeta_\beta} - \frac{\tau^\beta}{\tau^\eta}\frac{\partial f^\alpha}{\partial \eta}\right)\theta_\beta + \frac{1}{\tau^\beta}\frac{\partial \bar{f}_\alpha}{\partial \eta}.$$



Original source details: GELFAND-P000920

Uncertain reading. The first numerator of the lower line has a bar above f. No index is visible on f.



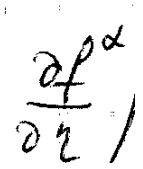
Original source details: GELFAND-P000920

Uncertain reading. The denominator after the minus sign tentatively has upper index η. The lower mark is heavily crossed out or overwritten.



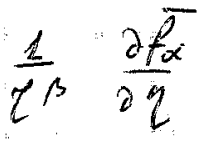
Original source details: GELFAND-P000920

Uncertain reading. A long looped cancellation lies between the fraction with τ^β and the next derivative.



Original source details: GELFAND-P000920

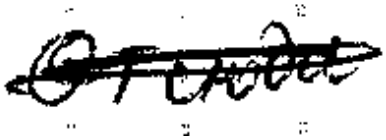
Uncertain reading. The middle numerator of the lower line has superscript α. No bar over f is visible.



Original source details: GELFAND-P000920

Uncertain reading. The last denominator reads τ^β, and the α on the last $\bar{f}$ is a subscript. No factor is visible after the derivative.

[Crossed out: From this.] Let



Original source details: GELFAND-P000920

Diagram and typed description. The crossed-out word reads ‘From this’.

$$\tau \mapsto s(z,\zeta)\tau(z,\zeta) \tag{19}$$

Then

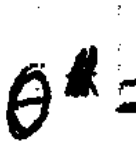
GELFAND-P000921

[source number: 3]

$$\begin{cases} \theta^* = s\theta \\ \theta^{\alpha *} = \theta^\alpha \\ \theta^*_\alpha = \theta_\alpha \end{cases} \tag{20}$$

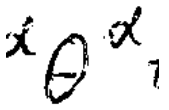
and

$$\begin{cases} \theta^* = u\theta^* \\ \theta^\alpha = u^\alpha \theta^\alpha + u^\alpha_\beta \theta^{\beta *} \\ \theta_\alpha = v_\alpha \theta^* + v^\beta_\alpha \theta^*_\beta \end{cases} \tag{21}$$



Original source details: GELFAND-P000921

Uncertain reading. On the left of the first row a heavily traced superscript over θ is tentatively read as a star.



Original source details: GELFAND-P000921

Uncertain reading. In the second line the first term on the right reads $u^\alpha\theta^\alpha$, with no visible star.

(22) A G-structure arises, where G is the group

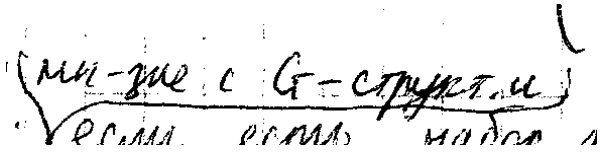
$$\begin{pmatrix} u & 0 & 0 \\ u^\alpha & u^\alpha_\beta & 0 \\ v_\alpha & 0 & v^\beta_\alpha \end{pmatrix}$$

(23) Abstr. Segre family: [crossed out above: collection?] [inserted above: (a set with a G-struct. and)] if there is a collection of forms.



Original source details: GELFAND-P000921

Uncertain reading. The word crossed out above the line is tentatively ‘collection’.



Original source details: GELFAND-P000921

Diagram and typed description. A long insertion mark leads to ‘if’. The word after ‘struct.’ reads ‘and’.

a set with a G-struct. and

and the conditions hold

$$\begin{cases} d\theta \equiv d\theta^{\alpha} \equiv 0 \pmod{\theta, \theta^1, \ldots, \theta^n} \\ d\theta \equiv d\theta_{\alpha} \equiv 0 \pmod{\theta, \theta_1, \ldots, \theta_n} \end{cases} \tag{24}$$

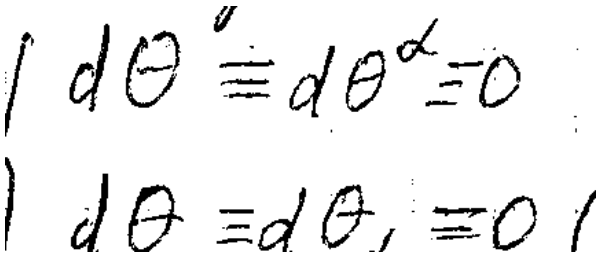


Diagram and typed description. Each of the four congruence signs in (24) has three strokes.

Original source details: GELFAND-P000921

Special case of the problem: what are the invariants of a family of curves in the plane, a family of surf. in 3-dimensional space, etc.

GELFAND-P000922

[source number: 6]

20.10.80 Gelfand Sem.

A. Goncharov. G-structures.

M — smooth manifold



Diagram and typed description. Bundle of tangent frames (principal $GL(n, R)$-bundle).

$$\begin{array}{c} E \\ \downarrow \\ M \end{array}$$

Original source details: GELFAND-P000922

$$G \subset GL(n, R)$$

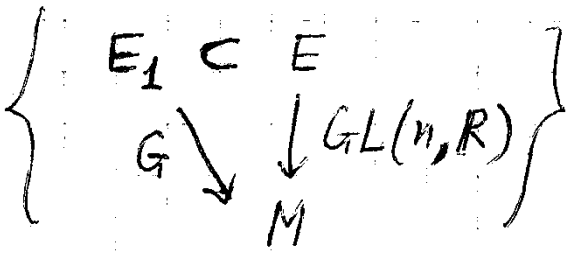


Diagram and typed description. G-structure.

$$\left\{ \begin{array}{ccc} E_1 & \subset & E \\ & \searrow^{G} & \downarrow GL(n,R) \\ & & M \end{array} \right\}$$

Original source details: GELFAND-P000922

Problem: [crossed out] consider all geometries on S^2 associated with the different subgroups $G \subset GL(n, R)$.



Uncertain reading. The word before 'consider' is crossed out and illegible.

Original source details: GELFAND-P000922

D. B. Fuchs. On cohomologies according to Deligne, MacPherson, etc.

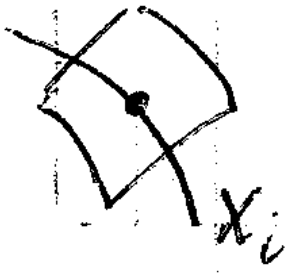
according to MacPherson.

X — set with sing., i.e. there is a stratification

$$X = X_n \supset X_{n-1} \supset \ldots \supset X_0. \tag{1}$$

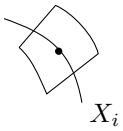
$X_i - X_{i-1}$ — smooth $2i$-dimensional set

and a certain natural condition of local triviality



Original source details: GELFAND-P000922

Diagram and typed description. Sketch accompanying the local triviality condition.



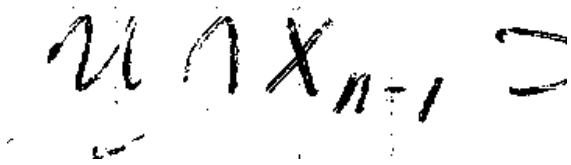
A point $x_0 \in X_i$ has a neighb. U such that

in the stratification

$$U \supset U \cap X_{n-1} \supset U \cap X_{n-1} \supset \ldots \supset U \cap X_i \tag{2}$$

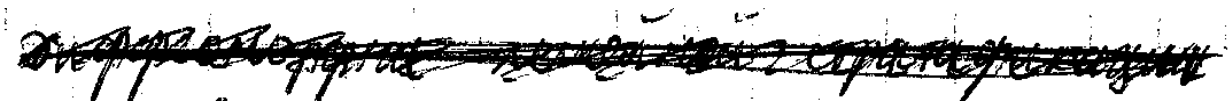
[crossed out: 'diffeomorphic' (?) [...] 'stratification' (?)]

has the same structure at every point of $U \cap X_i$



Original source details: GELFAND-P000922

Diagram and typed description. The term $U \cap X_{n-1}$ is written twice in succession in (2).



Original source details: GELFAND-P000922

Undeciphered passage. The middle of the crossed-out line is illegible. The beginning and ending are tentative readings.

Consider cycles of [circled and inserted above: sing.] simplices σ^q for which

$$\dim(\sigma^q \cap X_{n-i}) \leq q - i \qquad (\text{Codim } X_{n-i} = 2i\,!) \tag{3}$$

(like the condition that would hold in general pos. with respect to the distinguished part)

If the right side is q or $q - 2i$ (!), one gets ordinary cohomology.

Quotient by [crossed out above] the boundaries of chains subject to an analogous condition.



Original source details: GELFAND-P000922

Uncertain reading. The crossed-out sign before 'boundaries' is illegible.

Another defn., hypothetically equiv. to this one

Deligne wanted it to work over finite fields.

a more spec. reason: there is hope of proving a connection with L^2-cohomology

(L^2 [inserted below: -forms] on approaching the stratum)

A. Beilinson. The defn. above can be given a local form (pass to a complex of sheaves, its cohom. will be MacPherson cohom.).

GELFAND-P000923

Now the same using diff. operators

Embed

$$X \hookrightarrow \underset{\text{set [without singularities]}}{Y}$$

$$X_{\text{reg}} = X_n - X_{n-1} \hookrightarrow X \hookrightarrow Y. \tag{4}$$

On $X_n - X_{n-1}$ consider the triv. system of diff. eqs., i.e.

the sheaf $\mathcal{O}_{X_{\text{reg}}}$, the sheaf of fns., also a sheaf of modules over the ring of diff. op.

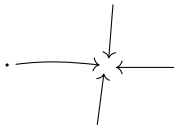
We want to extend this to all of Y.

Example:

X — a cross, Y — the plane



Diagram and typed description. Four arrows point inward. There is a separate dot to the left of the left arrow. Its role in the diagram is uncertain.



Original source details: GELFAND-P000923

extend the fns. by tensoring with δ in the transverse direction

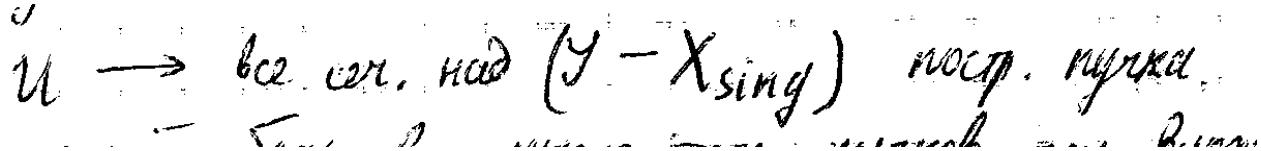
Diff. operators act on them.

(and the same in the general case)

Along X they must be reg. (and supported on X).

Consider the sheaf

$$U \longrightarrow \text{all sections over } (Y - X_{\text{sing}}) \text{ of the constructed sheaf.}$$



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Diagram and typed description. After 'over', the manuscript gives $(Y - X_{\text{sing}})$ without an intersection with U.

(5) (direct image in the sense of sheaf theory under the embedding $Y - X_{\text{sing}} \overset{i}{\hookrightarrow} Y$.)

(6) Thm. 1.

This sheaf of modules has a unique irred. submodule!

(Moral (I.M.G.) in the theory of sing. one should work with diff. operators.)

Thus a module over the ring of diff. op. over Y is constructed.

(sheaf of modules)

On X_{reg} consider the sheaf of fns. and extend them by δ-fns. in the transv. dir.

On $Y \setminus X_{\text{sing}}$ obtain a module over diff. op. (with holom. or algebr. coeffs.)



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Uncertain reading. The subscript on this line is overwritten. The final reading is ‘sing’.

Extend into the singularity X_{sing}.

The module is constr. in 2 steps

Step 1. Maximal — the direct image of the sheaf.

Step 2. Take the irred. submodule (it is unique!!)

Here: regular along X, but without compatibility!

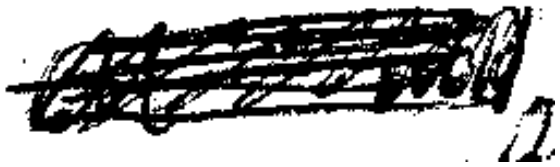


Original source details: GELFAND-P000923

Diagram and typed description. (4 different limits!)



[crossed out above] Consider a resolution of this module M

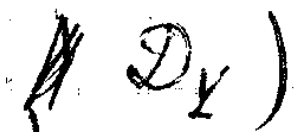


Original source details: GELFAND-P000923

Undeciphered passage. The word above ‘resolution’ is heavily crossed out and illegible.

$$\ldots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0 \tag{7}$$

(F_j — free module over diff. op. [crossed out] $\mathcal{D}_Y$)



Original source details: GELFAND-P000923

Uncertain reading. The cancelled sign before $\mathcal{D}_Y$ is illegible.

and map into fns. over Y (into $\mathcal{O}_Y$)

GELFAND-P000924

[source number: 7]

$$\bullet \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0$$

A complex arises:

$$\mathrm{Hom}_{\mathcal{D}}(F_0, \mathcal{O}_Y) \longrightarrow \mathrm{Hom}_{\mathcal{D}}(F_1, \mathcal{O}_Y) \longrightarrow \dots \tag{8}$$



Diagram and typed description. The abbreviation before ‘complex’ reads «Возн.», ‘arises’.

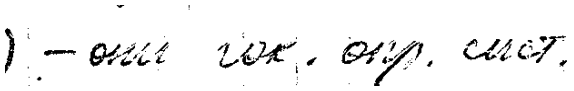
Original source details: GELFAND-P000924

up to $\mathrm{Codim}_Y X$ this is acyclic, then MacPherson cohomology

(9) The simplest overd. syst. of diff. eqs. — those for which there is locally a max. number of

(10) solutions (reduce to the system $\frac{\partial u}{\partial x_1} = 0, \dots, \frac{\partial u}{\partial x_n} = 0$) — they are wholly (?) determ. by the syst.

of their solns. — there will be a locally constant sheaf on the set.



Uncertain reading. The abbreviation after ‘they’ is tentatively read as «цел.» (‘wholly’).

Original source details: GELFAND-P000924

The solns. are locally const.

There may be no global solns.

Analogue in the general sit.

Consider any overd. syst.

(11) If the syst. has reg. singularities (there is a defn.), then the solutions are

— the MacPherson complex.

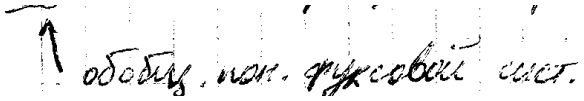


Diagram and typed description. Right-hand annotation. The arrow points to ‘reg. singularities’ in (11).

↑ generaliz. of the notion of a Fuchsian syst.

Original source details: GELFAND-P000924

This establ. a corresp. between these complexes and syst. with reg. sing.

Definition (Kashiwara) syst. with reg. sing.

Classical defn. (for syst. of ordinary eqs.) — reg. sing. (Fuchsian syst.):

$$\frac{dx}{dt} = \frac{1}{t}A(t)x, \quad \text{where } A \text{ is reg.} \tag{12}$$

(first-order syst.)

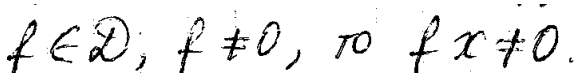
Let [crossed out] a $\mathcal{D}$-module M be given (assoc. with an overd. syst.)



Original source details: GELFAND-P000924

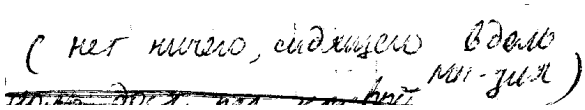
Uncertain reading. The writing crossed out before ‘given’ is illegible.

(13) a) Let M have no torsion, i.e. if $x \in M$, $x \neq 0$, $f \in \mathcal{D}$, $f \neq 0$, then $fx \neq 0$.



Original source details: GELFAND-P000924

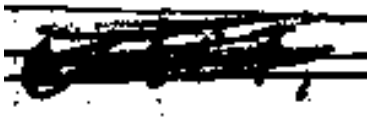
Diagram and typed description. The shape of the letter in $f \in \mathcal{D}$ matches $\mathcal{D}$ in the line about the module.



Original source details: GELFAND-P000924

Diagram and typed description. At right: (nothing is supported along sets)

[crossed out: Then [...] req. that the restr. along a suff. reg. curve ha[...] form (12)]



Original source details: GELFAND-P000924

Undeciphered passage. The heavily crossed-out word after ‘Then’ is illegible.

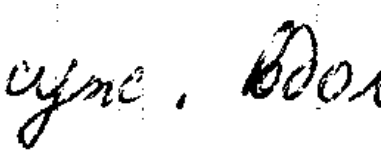


Original source details: GELFAND-P000924

Undeciphered passage. Before ‘form (12)’ the beginning «име-» (‘на-’) is visible.

(14) Then it reduces to syst. (10)

(15) Claim. M has reg. sing. $\Longleftrightarrow$ the restr. along a suff. reg. curve has form (12)



Original source details: GELFAND-P000924

Diagram and typed description. The abbreviation after the equivalence sign reads «суж.» (‘restr.’), as in the crossed-out line above.

General defn. In $\forall$ module there is a filtration
(finite length!) and the irred. factors has form (10) —
— integrable connection.



Original source details: GELFAND-P000924

Diagram and typed description. The line about the factors uses the singular verb ‘has’.

Defn. Each irred. compon. must have reg. sing.

GELFAND-P000925

27.10.80 Gelfand Sem.

Problems (for first-year students)

1. Find all subgroups and norm. subgroups of the symmetry group (?) of regular polyhedra and of the symmetric group.

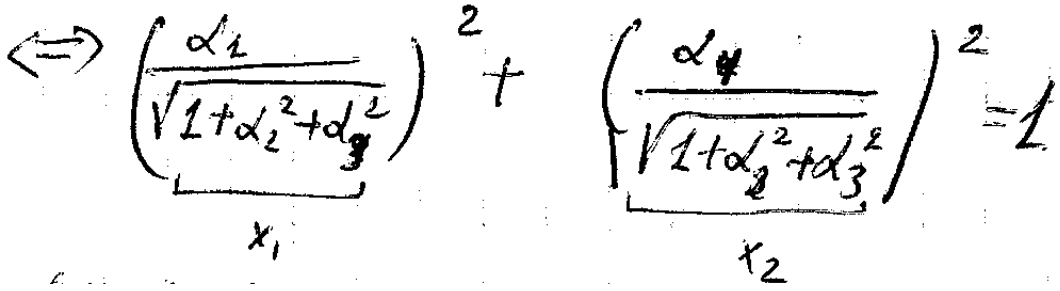


Uncertain reading. The ending of the word before ‘symmetry’ is heavily overwritten. The construction suggests the genitive «группы» (‘of the group’). The ending is illegible.

Original source details: GELFAND-P000925

2. Is the group $SL(2,R)$ simply connected? Construct its univ. cover.

$$\alpha\delta-\beta\gamma=1 \iff \alpha_1^2-\alpha_2^2-\alpha_3^2+\alpha_4^2=1$$
$$\iff \alpha_1^2+\alpha_4^2=1+\alpha_2^2+\alpha_3^2 \iff$$
$$\iff \left(\underbrace{\frac{\alpha_1}{\sqrt{1+\alpha_2^2+\alpha_3^2}}}_{x_1}\right)^2+\left(\underbrace{\frac{\alpha_4}{\sqrt{1+\alpha_2^2+\alpha_3^2}}}_{x_2}\right)^2=1.$$



Original source details: GELFAND-P000925

Uncertain reading. The subscripts 3 in the first denominator, 4 in the second numerator and 2 in the second denominator are overwritten. The original signs are illegible.

$$(\alpha_1,\alpha_2,\alpha_3,\alpha_4) \longmapsto (x_1,x_2)\in S^1$$
homotopy equiv.

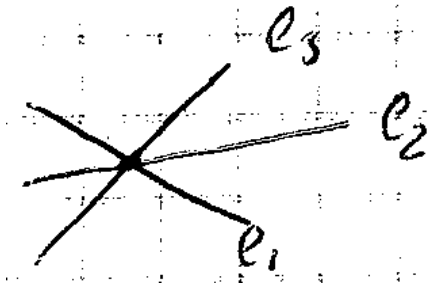
More precisely, one obtains

$$SL(2,R) \underset{\text{diffeom.}}{\cong} S^1\times R^2.$$

3. How is the univ. cover of $SL(2,R)$ structured as a group?

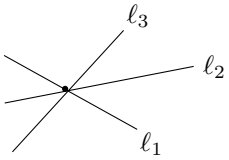
GELFAND-P000926

Suppose there are 3 lines

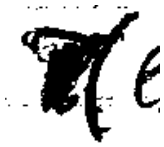


Original source details: GELFAND-P000926

Diagram and typed description. Three lines labelled ℓ_1, ℓ_2, ℓ_3.



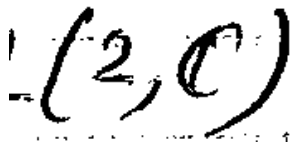
$$[?](\ell_1,\ell_2,\ell_3) = \begin{cases} +1 & \text{if } \ell_2 \text{ is inside } \widehat{\ell_1\ell_3} \text{ (angle } < \pi). \\ -1 & \text{otherwise} \end{cases}$$



Original source details: GELFAND-P000926

Uncertain reading. The sign before the arguments is heavily overwritten. Possible readings are τ and f. The right-hand note uses f for the three-argument function.

(g_j, n_j) — an element, $g_j \in SL(2,C)$

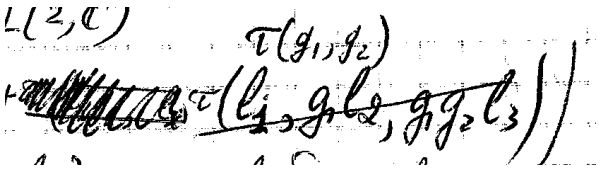


Original source details: GELFAND-P000926

Diagram and typed description. This line reads C. Question 3 on the preceding frame has R.

$$(g_1,n_1)(g_2,n_2) = (g_1g_2, n_1+n_2+\tau(g_1,g_2))$$

[crossed out $\tau(\ell_1, g_1\ell_2, g_1g_2\ell_3)$; inserted above the formula $\tau(g_1,g_2)$.]



Original source details: GELFAND-P000926

Undeciphered passage. A second, more heavily cancelled entry follows n_1+n_2+. Its reading is unresolved.

(ℓ_1, ℓ_2, ℓ_3) can be chosen arbitrarily, leading to a homolog. cocycle

$$g = \begin{pmatrix} \alpha & \beta \\ 0 & \alpha^{-1} \end{pmatrix} \qquad \varphi(g) = \begin{cases} 1, & \alpha > 0 \\ -1, & \alpha < 0 \end{cases}$$

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \varphi(g) = \begin{cases} i & c > 0 \\ -i & c < 0 \end{cases}$$

(if it does not reduce to triang.)

$$\varphi(g, n) = e^{i\frac{\pi}{2}n}\varphi(g)$$

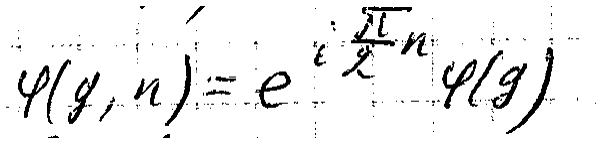


Diagram and typed description. The φ in the exponential formula has the same loop as in the two formulas above and in the definition of $\widetilde{G}$.

Original source details: GELFAND-P000926

Now take

$$\widetilde{G} = \{(g, n) : \varphi(g, n) = 1\}$$

$\widetilde{G}$ — this is a group and it is the univ. cover.

[at right] (Here $\ell_3 = \ell$ is the X-axis)

$$\tau(g_1, g_2) = f(\ell, g_1\ell\, g_1g_2\ell) = \pm 1$$



Uncertain reading. Between the second and third arguments a separate comma cannot be distinguished.

Original source details: GELFAND-P000926

ℓ — an oriented line

(invar. of a triple of [inserted below and circled: oriented] lines; index of a triple of lines;

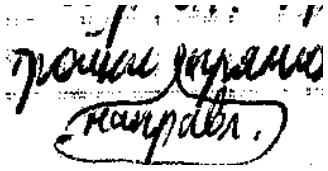


Diagram and typed description. The abbreviation for ‘oriented’ is circled and linked to the first occurrence of ‘lines’.

Original source details: GELFAND-P000926

$+1$ if $g_1\ell$ is between ℓ and $g_1g_2\ell$

This group is disconnected — there are 4 pieces; then take a conn. comp.

4. (Kirillov). Describe all locally projective structures on R

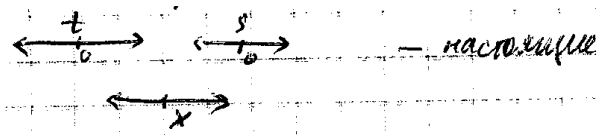
(covered by charts, transition — fractional-lin. transf.)

atlases are equiv. if the transition is a fract.-lin. transf.

Answer: there are countably many loc. project. structures

Example 1. R and param. t

Example 2:



Original source details: GELFAND-P000926

Diagram and typed description. Two upper axes t, s with zeros, and the lower axis x.

t (0) s (0) — genuine

x

Relations: $tx = -1, \quad xs = -1$

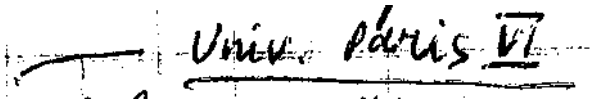
When one ends — go to the other, etc.

The second is not equiv. to the first.

A 2-sheeted cover of the project. line has a loc. project. structure.

A cover of the project. line — its project. transf. will be — transf. of the univ. cover.

Prof. Gaveau (from Paris) 3 possible topics.

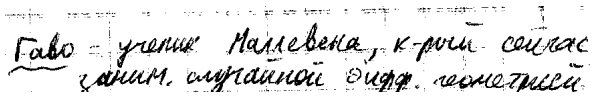


Original source details: GELFAND-P000926

Diagram and typed description. Annotation above the line with a leader to ‘from Paris’.

← Univ. Paris VI

1. Constr. of envelopes of holomorphy
2. Constr. of measures on certain ∞-dim. homog. sets
3. Scattering theory and functional integrals.



Original source details: GELFAND-P000926

Diagram and typed description. At right: Gaveau is a student of Malliavin, who is now working on stochastic diff. geometry.

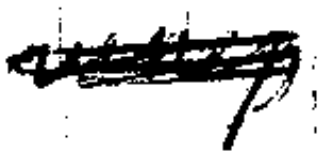
1. On constructions of envelopes of holomorphy.

Consider C^2 and try to constr. the envelope of holom. of a compact 2-dimensional surf. (in C^2)

Problem. Find such a [crossed out] Levi-flat hypersurf. M that

$$\partial M = S^2$$

(the Levi form on it must be $= 0$)

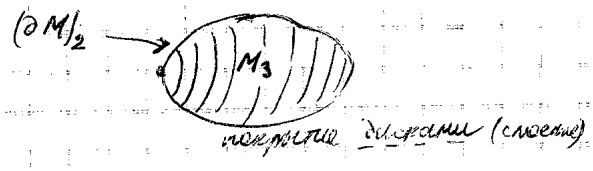


Original source details: GELFAND-P000926

Undeciphered passage. The word before ‘Levi-flat’ is heavily crossed out and illegible.

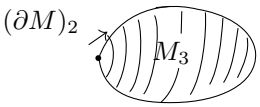
A Levi-flat region is foliated into compl. leaves

(On the torus this problem cannot be solved)



Original source details: GELFAND-P000926

Diagram and typed description. At the top of the oval is a short dark stroke whose direction is unclear.

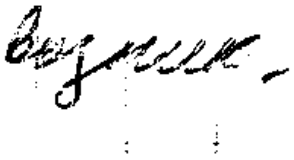


covering by discs (foliation)

GELFAND-P000927

[source number: 11]

There are sing. points of the foliation $\Rightarrow$ the index of the degen. (?) Riemann prob. $= 0 \Rightarrow$ it is solvable



Original source details: GELFAND-P000927

Uncertain reading. The abbreviation before 'Riemann prob.' is tentatively «вырожд.» or «вырож.» ('degen.').

(Bishop — there arises a sing. integro-diff. eq.)

2-3.

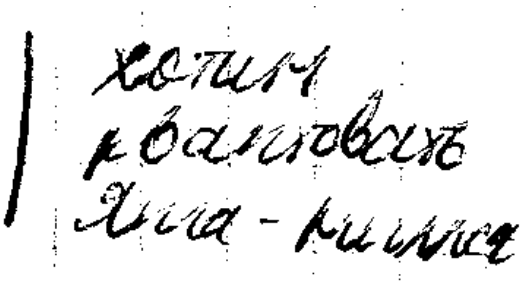
M — a set, G — a comp. Lie group; g — a Lie alg.

$\Lambda^1(M, g)$ — 1-forms on M with vals. in g.



Original source details: GELFAND-P000927

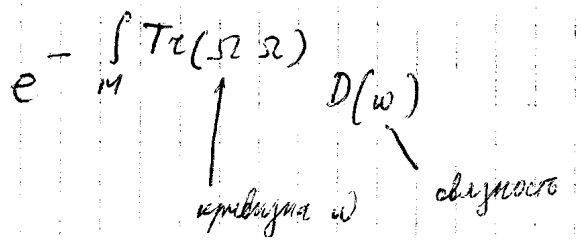
Uncertain reading. The subscript on Λ is crossed out. It may be the digit 2.



Original source details: GELFAND-P000927

Diagram and typed description. At right: 'we want to quantize Yang–Mills'.

$$e^{-\int_M \mathrm{Tr}(\Omega\Omega)} D(\omega) \tag{1}$$



Original source details: GELFAND-P000927

Diagram and typed description. Curvature and connection are labelled below (1).

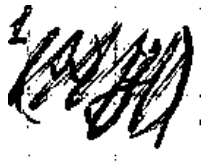
Ω ω

curvature of ωconnection

(Formal Feynman integral)

Group: $g \in C(M, G)$ (gauge)

$$g \cdot \omega = g^{-1} dg + g^{-1} \omega g \tag{2}$$



Original source details: GELFAND-P000927

Undeciphered passage. The expression crossed out immediately after the second g^{-1} is illegible.

Goal — construct a measure inv. under the gauge [crossed out] group.



Original source details: GELFAND-P000927

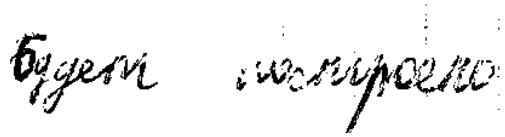
Uncertain reading. A short entry before ‘group’ is crossed out. It may read «гр.» (‘grp.’).

(easier than on the quotient space)

We must integrate over the set of paths in Λ^1.

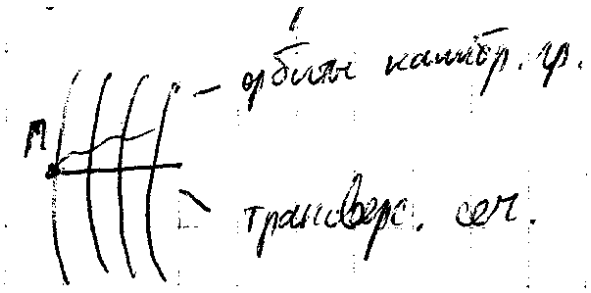
$$e^{-\int_0^t dt \int_M \mathrm{Tr}(\Omega_t \Omega_t)\, dx} D(\omega_t). \tag{3}$$

We will be constructed a family of measures, ‘approx.’ to the equil. measure.



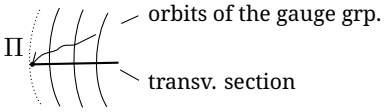
Original source details: GELFAND-P000927

Diagram and typed description. The manuscript reads «Будем построено», literally ‘we will be constructed’.



Original source details: GELFAND-P000927

Diagram and typed description. Orbits, the point Π and two lines issuing from it.

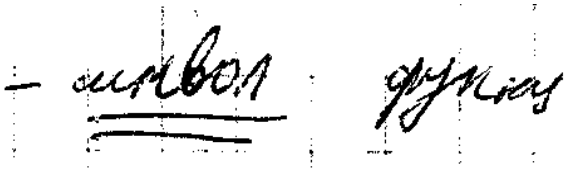


At first — a kind of Wiener measure on paths.

We need something like an ellipt. operator on the space of connections, inv. under the gauge group.

$$\Pi \to d_\Pi \text{ — covar. diff.}, \qquad \Box_\Pi = d_\Pi^* d_\Pi + d_\Pi d_\Pi^*$$

$$e^{-\tau \Box_\Pi}(x, y) = p_\tau^{(\Pi)}(x, y) \qquad \text{— symbol of the func.} \tag{4}$$



Original source details: GELFAND-P000927

Diagram and typed description. The word ‘symbol’ is double-underlined after (4).

$$L\varphi(\Pi) = \iint_{M\times M} e^{-\tau\Box_\Pi}(x,y)\frac{\delta^2\varphi}{\delta\Pi(x)\delta\Pi(y)}\,dx\,dy \tag{5}$$

[crossed out] τ fixed.



Original source details: GELFAND-P000927

Undeciphered passage. A sign or short word before τ is crossed out. The reading is unresolved.

One cannot work in Sob. spaces.

$$\Lambda^1_{k,\alpha}(M,g), \quad 0<\alpha<1, \tag{6}$$

— some space (Hölder)

The kth deriv. is Hölder with exponent α.

We need:

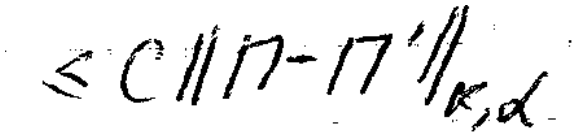
$$\left\|e^{-\tau\Box_\Pi} - e^{-\tau\Box_{\Pi'}}\right\|_{HS} \le C\|\Pi-\Pi'\| \tag{7}$$

[under HS: Hilb.–Schmidt.]

GELFAND-P000928

One obtains

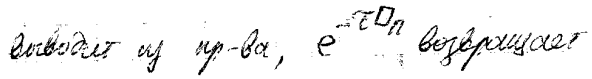
$$\left\|e^{-\tau\Box_\Pi} - e^{-\tau\Box_{\Pi'}}\right\|_{HS(L^2,L^2)} \le C\|\Pi-\Pi'\|_{k,\alpha} \tag{8}$$



Original source details: GELFAND-P000928

Diagram and typed description. In (8) there is a slanting stroke beneath $<$. The sign reads $\le$.

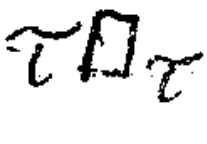
Techn. point: Brownian motion takes us out of the space, $e^{-\tau\Box_\Pi}$ brings us back.



Original source details: GELFAND-P000928

Diagram and typed description. Before the exponential the manuscript reads ‘takes us out of the space’.

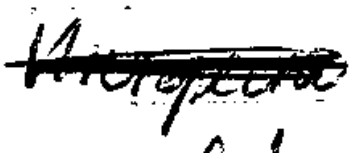
$$d\Pi_t = e^{-\tau\Box_\tau}\underbrace{d\beta_t}_{\text{white noise}}. \tag{9}$$



Original source details: GELFAND-P000928

Diagram and typed description. In (9) the subscript on $\Box$ has the same shape as the preceding τ.

[crossed out: ~~Interesting~~]



Original source details: GELFAND-P000928

Diagram and typed description. The word ‘Interesting’ is crossed out above (10).

$$\left(\tfrac{1}{2}\Delta - V\right)\Psi = \frac{\partial}{\partial t}\Psi \tag{10}$$

$$\Psi_t(x) = E_x\left(\exp\left(-\int_0^t V(b_\omega(s))\,ds\right)\Psi_0(b_\omega(t))\right) \tag{11}$$

In particular,

$$\pm\lambda_1 = \lim_{t\to\infty}\frac{\log E_x\left(\exp\left(-\int_0^t V(b_\omega(s))\,ds\right)\right.}{t} \tag{12}$$

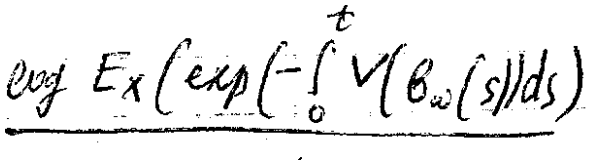


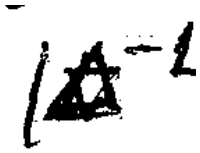
Diagram and typed description. The numerator of (12) has just one closing parenthesis after ds.

Original source details: GELFAND-P000928

(Interesting, e.g. when there are neg. levels)

Suppose

$$\left|\Delta^{-1}V\right| \le C < 1 \tag{13}$$

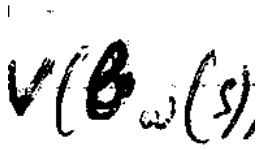


Uncertain reading. The sign before $^{-1}$ is overwritten. The outer contour reads as Δ. The earlier sign is unresolved.

Original source details: GELFAND-P000928

Then

$$E_x\left(\exp\left(-\int_0^\infty V(b_\omega(s))\,ds\right)\right) \le \frac{1}{1-C} \tag{14}$$



Uncertain reading. The letter in (14) is overwritten. The reading b_ω follows (11)–(12), with B_ω also possible.

Original source details: GELFAND-P000928

$\Rightarrow$ estimate the eigenvalue.

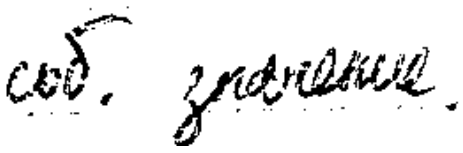
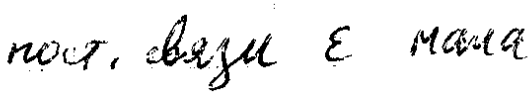


Diagram and typed description. The words ‘соб. значение’ read ‘eigenvalue’.

Original source details: GELFAND-P000928

$V \longmapsto \varepsilon V$ pot. (?) we took ε small $\Rightarrow$ something is estimated.



Uncertain reading. The abbreviation after εV is unclear. It may be ‘pot.’ (potential).

Original source details: GELFAND-P000928

GELFAND-P000929

3/XI-80 Gelfand Sem.

I.M.G. History of representation theory

1. Dedekind — characters of finite commut. groups (for L-fns.)
2. Frobenius — matrices and traces for noncommut. groups
3. Haar (died young)

(*)4. von Neumann — theory of C^*-algebras for repns. of noncommut. groups

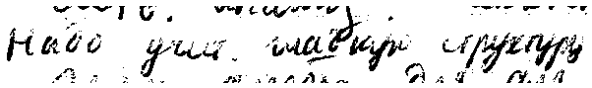
(**)5. Gelfand and Naimark — on repns. of the Lorentz grp.

Ideology (*): everything is set up in set-theoretic terms

(**). —//— smoothly and simply

Modern analysis: noncommut. diff. geometry and integral geometry.

One must take account of the smooth structure.



Original source details: GELFAND-P000929

Diagram and typed description. In the cramped line after ‘take account of’, the adjective reads ‘smooth’. A short stroke underlines its beginning.

Group algebra for a Lie alg. — enveloping algebra.

but there are still many of them (no Wedderburn thm.)



Original source details: GELFAND-P000929

Diagram and typed description. The manuscript spells the surname «Веддербарна».

Gelfand, Kirillov, skew field of fractions of the enveloping alg. ≅ skew field of fractions of the ring of diff. operators

This is unfinished. (Operators with polyn. coefficients are insufficient. This was partly understood in the work of [crossed out] Novikov, Krichever, Shabat, etc.)



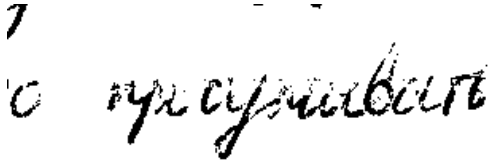
Original source details: GELFAND-P000929

Uncertain reading. The entry before ‘Novikov’ is heavily crossed out and illegible.

Bernstein, S. Gelfand, I.M. Gelfand in project. space.

One must work with sheaves.

Now one must listen to (?) what representation theory is being developed for, rather than to the theory itself.



Original source details: GELFAND-P000929

Uncertain reading. The opening strokes of the verb admit «прислушивать» and «прослушивать», both forms of ‘listen’.

GELFAND-P000930

[source number: 16]

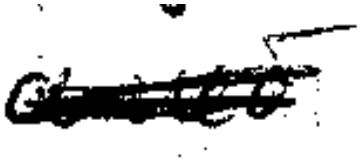
A. Beilinson

X — an algebr. set over $\mathbb{C}$

The simplest — affine.

They are uniquely determ. by their ring of fns.

Commut. ring = [crossed out: algeb. (?)] affine alg. set.



Original source details: GELFAND-P000930

Uncertain reading. The crossed-out entry after the equals sign is tentatively read «алгеб.» (‘algeb.’).

Modules over it = quasicoher. sheaves

$X = \operatorname{Spec} A$, M — an A-module (always the Zariski top.)

basis of open sets: $X_f = \{x : f(x) \neq 0\}$.

$O(X_f) = A_f$ — the localization of A at f

$$\mathcal{M}(X_f) = M_f = M \otimes_A A_f$$

We obtain a sheaf on Spec A. This gives the class of quasicoher. sheaves on X.

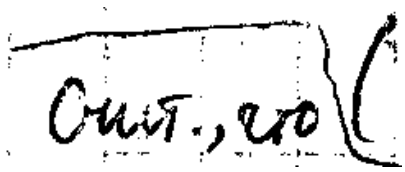
There will be a noncommut. analogue of localization. (passage from a mod. to a sheaf).

(1) Rings: univ. enveloping algs. of semisimple Lie algs. and their quotients.

X — an alg. set — is called $\mathcal{D}$-affine, if it has the property

($\mathcal{D}_X$ — the sheaf of rings of diff. operators on X)

X — smooth;



Original source details: GELFAND-P000930

Diagram and typed description. At left beside the smoothness line: ‘Ass., that’.

$\mathcal{M}$ — a sheaf of modules over $\mathcal{D}_X$, $O_X \subset \mathcal{D}_X$ (subsheaf, diff. operators of order 0)
$\mathcal{M}$ is called quasicoherent, if it is quasicoher. as a sheaf of O_X-modules.
(Example: $\mathcal{D}_X$ itself is quasicoherent).
X — an algebr. set, O_X — the sheaf of fns. on it; a sheaf $\mathcal{M}$ of O_X-modules on X
is called quasicoherent, if it has the following property:
$\forall$ affine set U and its aff. subset $U' \subset U$ there must be:

$$\mathcal{M}(U') = \mathcal{M}(U) \otimes_{O(U)} O(U') \tag{2}$$

(the right side of (2) clearly maps to the left, and we require this to be an isomorphism)
(3) Assertion. X is affine $\Longleftrightarrow \forall$ quasicoher. $\mathcal{M}$
(4) $H^i(X, \mathcal{M}) = 0$ and $\mathcal{M}$ is gen. over $O(X)$ by global sections, i.e.

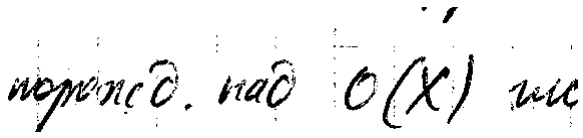


Diagram and typed description. In (4) the manuscript has $O(X)$. In the following line the tensor sign has no ring subscript.

Original source details: GELFAND-P000930

$$H^0(X, \mathcal{M}) \otimes O_X \underset{\text{epimorphism}}{\longrightarrow} \mathcal{M} \tag{5}$$

(A theorem that is not quite triv. — Serre, Grothendieck — see Serre's work on coher. sheaves)

GELFAND-P000931

(6). Definition. X is called $\mathcal{D}$-affine, if $\forall$ quasicoher. sheaf of $\mathcal{D}$-modules $\mathcal{M}$ the conditions (4), (5) hold, i.e.



Undeciphered passage. An earlier label is crossed out to the left of (7). It resembles (8), but the reading is unresolved.

Original source details: GELFAND-P000931

$$H^i(X, \mathcal{M}) = 0, \quad i > 0 \tag{7}$$

$$H^0(X, \mathcal{M}) \otimes \mathcal{D}_X \longrightarrow \mathcal{M} \longrightarrow 0 \tag{8}$$

(9) Examples.

1. $\forall$ aff. set — $\mathcal{D}$-affine.

2. $\forall$ project. space $\mathbb{P}^N$ — $\mathcal{D}$-affine.

(partic. case of the thm. to be discussed)

3. An elliptic curve — is not $\mathcal{D}$-affine.

(10) Consequence of $\mathcal{D}$-affinity

Let X be a set, $\mathcal{L}$ a sheaf of O-modules. Consider

(11) $\mathcal{D}_X \otimes_{O_X} \mathcal{L}$ — a quasicoher. sheaf of $\mathcal{D}_X$-modules

[crossed out above] (12) X — $\mathcal{D}$-aff. $\Rightarrow$ the cohomology of sheaf (11) is 0 and it is gen. by global sects. (i.e. (7) and (8) hold), even if $\mathcal{L}$ itself was bad



Undeciphered passage. A short word is crossed out above the start of (12). Its reading is unresolved.

Original source details: GELFAND-P000931

(13) (e.g., $O(-i)$ on $\mathbb{P}^N$, it has no sections and has higher cohomology)

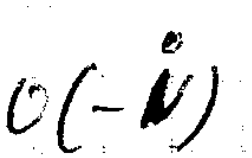


Diagram and typed description. A dot is visible above the heavily traced letter. It reads i.

Original source details: GELFAND-P000931

e.g.,

(14) $\mathcal{D}_X \otimes_{O_X} O(-i)$ on P^N is gen. by global sections

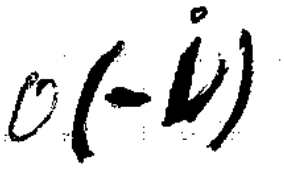


Diagram and typed description. In (14), the notation is again $O(-i)$, with i traced over.

Original source details: GELFAND-P000931

(the speaker does not know a direct proof)

Unfortunately there is no good notation for $\mathcal{D}_X$, since a diff. operator is not written globally as a sum of homogeneous ones.

(14) — a certain loc. free $\mathcal{D}_X$-module.

$\mathcal{D}_X$ in (11) is regarded as a right sheaf of O_X-modules



Uncertain reading. The last digit in the reference to (11) is heavily traced over.

Original source details: GELFAND-P000931

If we consider

$$O(-i) \otimes_{O_X} \mathcal{D}_X, \tag{15}$$

then this is no longer so. It is not gen. by global sects. and its higher cohom. is $\neq 0$ ($O(-i)$ is its direct summand)

[crossed out: ~~Elliptic curve~~] On the elliptic curve.

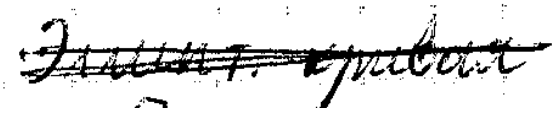


Diagram and typed description. The earlier heading is crossed out before the new one.

Original source details: GELFAND-P000931

(16) Suppose $\mathcal{D}_X$ has an involution, ident. on O_X, and $ab \overset{*}{\mapsto} ba$;

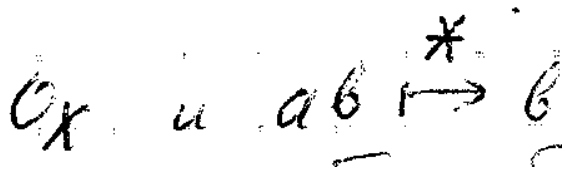


Diagram and typed description. An asterisk is written above the arrow from ab to ba.

Original source details: GELFAND-P000931

then (14) and (15) would be isomorphic and the set would not be $\mathcal{D}$-aff.

But on the ell. curve there is an involution — taking the formally adj. operator

$$f \cdot \frac{\partial}{\partial x} \mapsto \frac{\partial}{\partial x} f$$

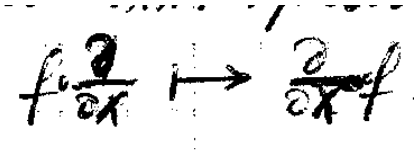


Diagram and typed description. There is no visible minus sign before the fraction on the right.

Original source details: GELFAND-P000931

(there are coords. — all operators are 'with const. coeffs.')

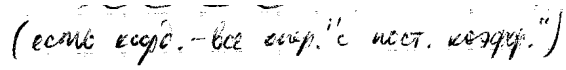


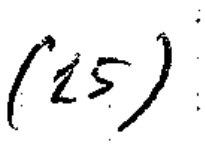
Diagram and typed description. The parenthesis opens with 'there are coords.'.

Original source details: GELFAND-P000931

GELFAND-P000932

[source number: 17]

The ring of fns. splits off from $\mathcal{D}_X \Rightarrow$ (15) is not acyclic



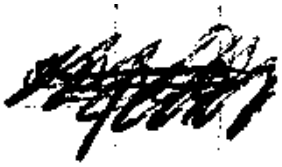
Uncertain reading. The first digit of the reference is overwritten. The link to the preceding frame supports reading (15).

Original source details: GELFAND-P000932

(17) Assertion. X — $\mathcal{D}$-affine, then [crossed-out writing unresolved] the section functor Γ

$$\mathcal{D}_X\text{-mod} \xrightarrow{\Gamma} \Gamma(\mathcal{D}_X)\text{-mod}, \tag{18}$$

is an equivalence of categories.



Original source details: GELFAND-P000932

Undeciphered passage. One crossed-out word between “then” and “the functor” is unresolved.

(The speaker does not know whether $\mathcal{D}$-affinity follows from this; it does in the commut. case)



Original source details: GELFAND-P000932

Uncertain reading. There is a small heavily inked mark after “The speaker”. Its function is unresolved.

$$D_X \underset{\text{def}}{=} \Gamma(\mathcal{D}_X) \tag{19}$$

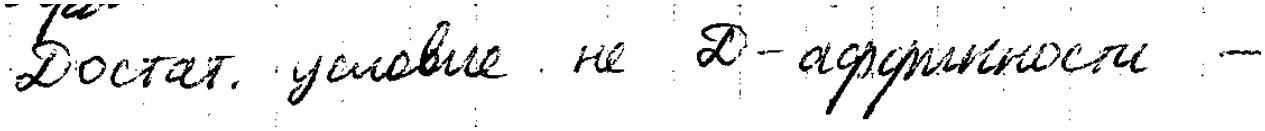
[crossed out: ~~Theorem~~ (?)]



Original source details: GELFAND-P000932

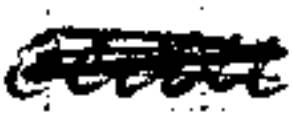
Uncertain reading. Crossed-out writing before (20) resembles “Theorem”.

(20) Suff. condition for failure of $\mathcal{D}$-affinity — [crossed-out writing unresolved] ch_1 (of the sheaf of forms of max. degree) $= 0$



Original source details: GELFAND-P000932

Diagram and typed description. No hyphen is visible between the Russian negation and $\mathcal{D}$ in the heading.



Original source details: GELFAND-P000932

Undeciphered passage. The crossed-out writing before ch_1 is unresolved.

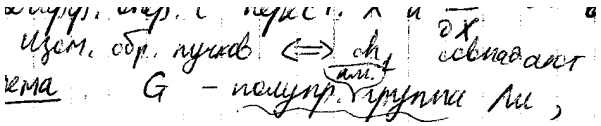
$$(\Leftrightarrow ch_1(\text{tangent bundle}) = 0) \tag{21}$$

$$(20) \Leftrightarrow (21) \Leftrightarrow \text{existence of an involution.} \tag{22}$$

(Diff. op. with x and $\frac{\partial}{\partial x}$ interchanged is a diff. op. on forms of max degree.)

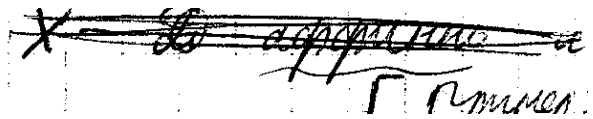
[interlinear insertion:] Isom. of inv. sheaves $\Leftrightarrow ch_1$ coincide

(23) Theorem. G — semisimple [inserted above and circled: alg. (?)] Lie group, B — Borel subgr., consid. $X = G/B$ — («flag space»). then [crossed out: ~~X — $\mathcal{D}$-affine and~~]



Original source details: GELFAND-P000932

Uncertain reading. The circled insertion resembles “alg.”. A leader descends toward “Lie group”. The exact reading is unresolved.



Original source details: GELFAND-P000932

Diagram and typed description. “X is $\mathcal{D}$-affine and” is crossed out after “Then”.

1. X — $\mathcal{D}$-affine
2. $D_X = U(\mathfrak{g})/U(\mathfrak{g})I_0$;

[in the right margin, with a square bracket at left]

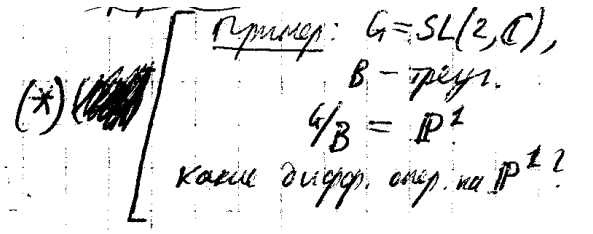
(*) ([crossed-out writing unresolved])

Example: $G = SL(2, \mathbb{C})$,

B — triang.

$G/B = \mathbb{P}^1$

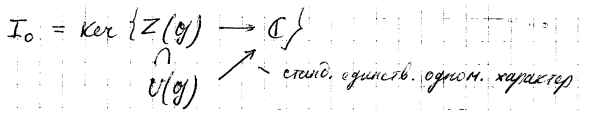
which diff. op. are there on $\mathbb{P}^1$?



Original source details: GELFAND-P000932

Undeciphered passage. The parenthesized writing before “Example” is crossed out. The exact reading is unresolved.

where



Original source details: GELFAND-P000932

Diagram and typed description. Diagram (24). The mark between $Z(\mathfrak{g})$ and $U(\mathfrak{g})$ resembles $\cap$.

$$I_0 = \ker\{\, Z(\mathfrak{g}) \longrightarrow \mathbb{C}\}$$
$$\cap \quad \nearrow$$
$$U(\mathfrak{g})$$

(24) — stand. unique one-dim. character

I_0 — Laplace operators without constant term

[crossed out: ~~Assertion~~ (?)] Proposition.



Original source details: GELFAND-P000932

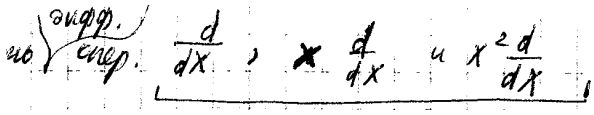
Uncertain reading. The crossed-out word before “Proposition” resembles “Assertion”.

(25) D_X-modules $\Leftrightarrow$ representations $\mathfrak{g}$, on which there acts trivially I_0.

($Z(\mathfrak{g})$ must act as on the triv. one-dimensional representation)

Example (*).

(26) On $\mathbb{P}^1$ there are [inserted above: diff.] op.



Original source details: GELFAND-P000932

Diagram and typed description. A bracket underneath groups the three operators in (26). A curved leader joins the inserted “diff.” to “op.”.

$$\frac{d}{dx}, \quad x\frac{d}{dx} \quad \text{and} \quad x^2\frac{d}{dx}$$

Take polynomials in them. This is the enveloping alg.

There is a unique op. commuting with all of them. (Shvartsman?)



Original source details: GELFAND-P000932

Uncertain reading. The surname is tentatively read as Shvartsman. The question mark is in the manuscript.

Moral: representation theory reduces to the theory of $\mathcal{D}_X$-modules.

Improvement over what was known: one can localize!!

GELFAND-P000933

10.11.80 Gelfand Sem.



Original source details: GELFAND-P000933

Uncertain reading. There are several isolated strokes above the date. No letters or digits can be distinguished.

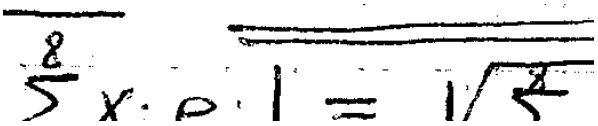
I. M. G. Cayley numbers.

$$\left|\sum_{j=1}^{8} x_j e_j\right| = \sqrt{\sum_{1}^{8} x_j^2} \tag{1}$$



Original source details: GELFAND-P000933

Uncertain reading. A heavily overwritten or crossed-out stroke appears to the left of the digit in (1). The earlier mark is unresolved.



Original source details: GELFAND-P000933

Uncertain reading. The upper limit under the radical touches its bar. It is tentatively read as 8 by comparison with the left side.

In the Cayley numbers:

$$|ab| = |a||b| \tag{2}$$

How to prove there are no > 8 dim.?

From (2) one gets a problem in repr. theory and an explicit construction.

Question: perhaps it exists in an ∞-dimensional space? (with cond. (2))

Gindikin asserts that it does. We must repeat the proof that in the finite-dimensional case there are none besides $\mathbb{R}, \mathbb{C}, \mathbb{H}$ and Cayley.

Projective geometries (planes).

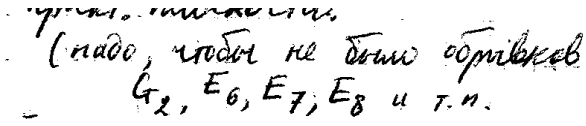
Introduce coord., associativity $\Leftrightarrow$ Desarguesian property

The Cayley proj. plane is the simplest non-Desarguesian one

It is interesting to study differ. and locally homogeneous proj. planes.

— The prob. of classifying Lie algebras is too particular (there should be no fragments

G_2, E_6, E_7, E_8 etc.



Original source details: GELFAND-P000933

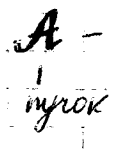
Diagram and typed description. The list of types is written below "there should be no fragments". No closing parenthesis is visible.

Beilinson. On localization of representations.

X — alg. variety (over $\mathbb{C}$),

$\mathcal{A}$ — quasi-coherent O_X-module

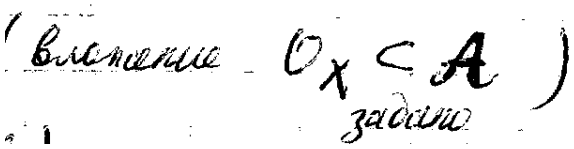
| a sheaf of rings containing O_X.



Original source details: GELFAND-P000933

Diagram and typed description. A short vertical stroke descends from $\mathcal{A}$ to the next line.

(the inclusion $O_X \subset \mathcal{A}$ [inserted below: is given])



Original source details: GELFAND-P000933

Diagram and typed description. "Is given" is inserted beneath the inclusion sign.

(E.g., $\mathcal{A}$ is a sheaf of diff. operators)

$\mathcal{A}$-mod — the category of sheaves of left $\mathcal{A}$-mod., that are quasi-coherent O_X-modules

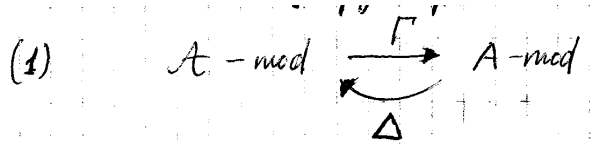
GELFAND-P000934

[source number: 20]

(Last time $\mathcal{A}$ was diff. op.)

$A = \Gamma(\mathcal{A})$, A-mod — categ. of A-modules (left).

There are 2 adj. functors.



Original source details: GELFAND-P000934

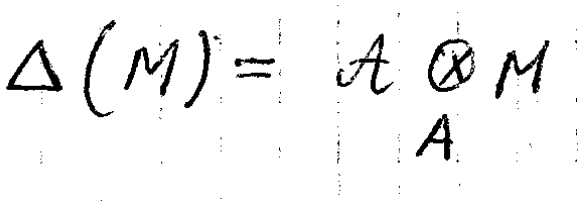
Diagram and typed description. In (1) the Γ arrow points right, with the curved Δ arrow pointing left below it.

$$\mathcal{A}\text{-mod} \underset{\Delta}{\overset{\Gamma}{\rightleftarrows}} A\text{-mod}$$

(1)

$$\Delta(M) = \mathcal{A} \underset{A}{\otimes} M, \tag{2}$$

M is consid. as a const. sheaf of $\mathcal{A}$-mod. over X (stalk $= M$ at each point) and the maps are trivial.



Original source details: GELFAND-P000934

Diagram and typed description. In (2) A is written directly beneath the tensor sign.



Original source details: GELFAND-P000934

Diagram and typed description. A calligraphic $\mathcal{A}$ precedes "mod. over X", as on the left in (1).

It is easy to see that they are adjoint.

(3) Defn.-lemma. [crossed out: ~~a~~ (?)]



Original source details: GELFAND-P000934

Uncertain reading. The sign after the heading is crossed out. It may be "a)".

A variety X is called $\mathcal{A}$-affine, if one of the 2 equiv. conditions holds

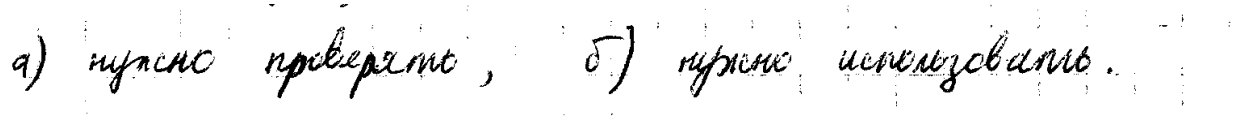
a) $\forall\, \mathcal{M} \in \mathcal{A}\text{-mod}, \quad H^i(X, \mathcal{M}) = 0, \quad i > 0$ and

$$H^0(X, \mathcal{M}) \neq 0$$
when $\mathcal{M} \neq 0$

b) The functors Γ and Δ are mutually inverse equivalences of categories.

(In fact a) is equiv. to what we had last time, we change the modules)

a) is what we need to check, b) is what we need to use.



Original source details: GELFAND-P000934

Diagram and typed description. The same word, “need”, is written before both “check” and “use”.

Proof of the equivalence of a) and b).

a) $\Rightarrow$ b).

$$\Gamma(\mathcal{M}) = \mathrm{Hom}_{\mathcal{A}\text{-mod}}(\mathcal{A}, \mathcal{M}) \tag{4}$$

(the image of 1 is a section of $\mathcal{M}$).

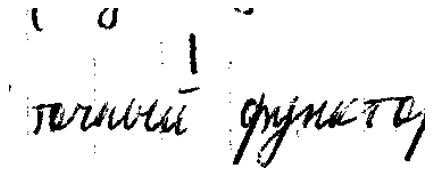
(5) a) $\Rightarrow \mathcal{A}$ — a projective generator in the categ. $\mathcal{A}$-mod.

(6) (an object such that i) $\mathrm{Hom}(\mathcal{A}, \mathcal{M})$ — exact functor

(7) ii) $\forall$ nonzero object $\mathcal{M}$; $\mathrm{Hom}(\mathcal{A}, \mathcal{M}) \neq 0$)

(i) follows from acyclicity, and ii) from the 2nd condition)

(exact functor — takes exact seq. to exact ones)



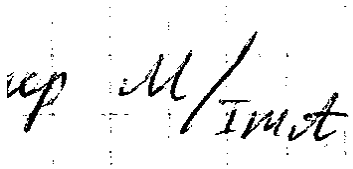
Original source details: GELFAND-P000934

Uncertain reading. A detached vertical stroke stands above the end of “exact”. It may be an exclamation mark.

i) — the property of projectivity

ii) — generator

(by the way, ii) $\Rightarrow$ we can map $\mathcal{A}$ to the quotient $\mathcal{M}/\operatorname{Im} t$ and so on)

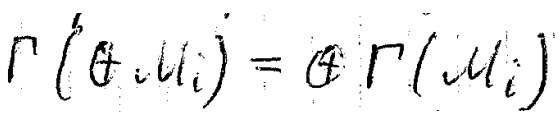


Original source details: GELFAND-P000934

Diagram and typed description. The symbol after Im in the denominator reads t.

(8) iii) Γ takes direct sums to direct sums.

$$\Gamma(\oplus_i \mathcal{M}_i) = \oplus \Gamma(\mathcal{M}_i)$$



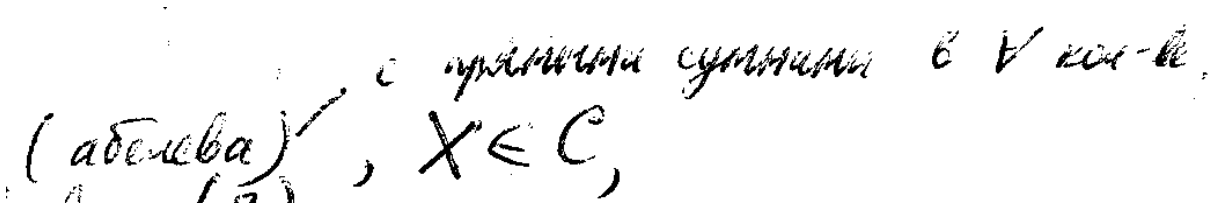
Original source details: GELFAND-P000934

Uncertain reading. A small stroke beside the left direct-sum sign is tentatively read as an index i. There is no visible counterpart at the right sign.

(a property of all quasi-coh. sheaves)

General categorical fact

(9) Lemma. Let there be given ∀ category C (abelian) [inserted above: with direct sums in ∀ quantity.], $X \in C$,



Original source details: GELFAND-P000935

Diagram and typed description. A leader from "(abelian)" points to the insertion about direct sums.

X — Proj. generator with property (8):



Uncertain reading. The X at the start of the line is heavily retraced. The writing underneath is unresolved.

Original source details: GELFAND-P000935

(10) $\mathrm{Hom}_C(X, \cdot)$ commutes with direct sums.

(11) [crossed out: ~~Then~~]

$$\mathrm{Hom}_C(X, \cdot) : C \longrightarrow \mathrm{End}_C X\text{-mod}$$

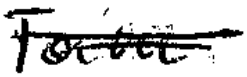


Diagram and typed description. "Then" is crossed out before formula (11).

Original source details: GELFAND-P000935

Then this functor is an equiv. of categories.

Everything follows from the lemma if we apply it to the case where

$$C = \mathcal{A}\text{-mod}, \quad X = \mathcal{A}, \quad \mathrm{End}_C X = A \tag{12}$$

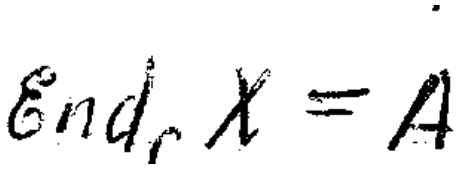


Diagram and typed description. The right side of (12) reads $\mathrm{End}_C X = A$.

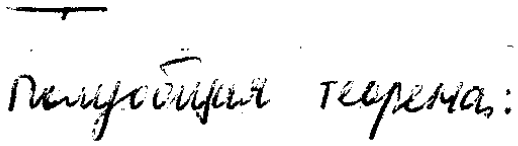
Original source details: GELFAND-P000935

The conv. implication b) $\Rightarrow$ a) from the Čech resolution.

$\mathcal{D}$-affine variety, a variety that is $\mathcal{A}$-aff. for $\mathcal{A} = \mathcal{D}_X$.

(13) Thm. $\mathbb{P}^N$ — $\mathcal{D}$-affine.

The following theorem:



Original source details: GELFAND-P000935

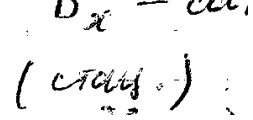
Diagram and typed description. The heading begins with “Следующая”, meaning “The following”.

G — a reductive group.

$X = G/B$ — flag space

(B — Borel subgroup)

(14) $x \in X \Rightarrow \underset{\text{(stab.)}}{B_x}$ — the corresp. Borel subgroup

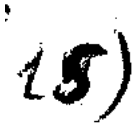


Original source details: GELFAND-P000935

Diagram and typed description. “(stab.)” is inserted beneath B_x.

G — acts on $X \Rightarrow \mathfrak{g}$ acts on X.

$$\mathfrak{g} \longrightarrow \text{Vect}\, X \tag{15}$$

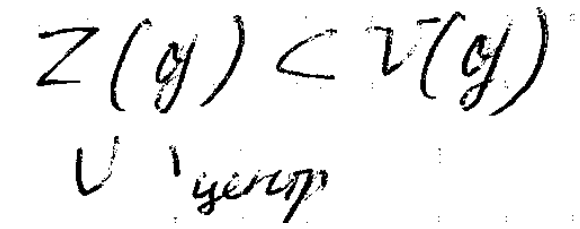


Original source details: GELFAND-P000935

Uncertain reading. The second digit in (15) is heavily retraced and reads 5.

hence.

$$U(\mathfrak{g}) \longrightarrow D_X = \Gamma(\mathcal{D}_X). \tag{16}$$



Original source details: GELFAND-P000935

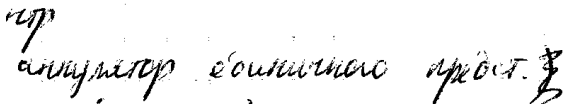
Diagram and typed description. A short hooked stroke under $Z(\mathfrak{g})$ stands next to the label “center”.

$Z(\mathfrak{g}) \subset U(\mathfrak{g})$

$\cup$ ' center

(17)

I — the annihilator of the unit repr. [?]



Original source details: GELFAND-P000935

Uncertain reading. The last sign after “repr.” is overwritten. The exact reading is unresolved.

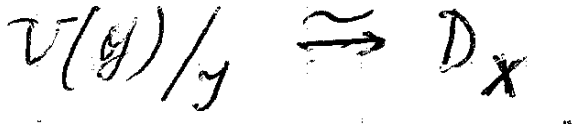
(max ideal in the center)

$$J = I \cdot U(\mathfrak{g}). \tag{18}$$

(19) Theorem. a) X — $\mathcal{D}$-affine.

b) under the morphism (16) one obtains an isomorphism

$$U(\mathfrak{g})/J \xrightarrow{\sim} D_X \tag{20}$$



Original source details: GELFAND-P000935

Diagram and typed description. In (20) a single arrow has a tilde above it.

(21) hence, equiv. of the categories of 1) $\mathcal{D}$-sheaves on X and 2) representations of the Lie alg. $\mathfrak{g}$ with unit infinitesim. character.

GELFAND-P000936

[source number: 21]

[crossed out: ~~Definition~~ (?)] Digression.



Original source details: GELFAND-P000936

Uncertain reading. The crossed-out heading resembles “Definition”. The exact reading is unresolved.



Original source details: GELFAND-P000936

Diagram and typed description. The new heading reads “Отступление”, or “Digression”.

X, Y — both smooth alg. varieties.



Original source details: GELFAND-P000936

Diagram and typed description. The word after X, Y reads “both”.

$X \xrightarrow{f} Y$ a map.

Then there are the categories $\mathcal{D}_X$-mod and $\mathcal{D}_Y$-mod

There are 2 functors on sheaves f^* and f_*.

But in diff. op. there is no corresp. morphism.

But the (natural) functors do exist, they are def. at the level of the derived category (actually there are 4 functors)

(22) f° — inverse image in the cat. of sheaves.

$$O_X \underset{f^\circ O_Y}{\otimes} f^\circ(\mathcal{D}_Y) \tag{23}$$

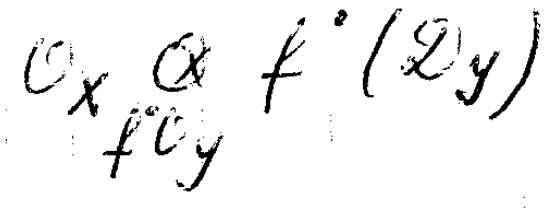


Diagram and typed description. In (23) the marks above f look like small hollow circles, as in (22).

Original source details: GELFAND-P000936

$$\mathcal{D}_{X\to Y} \underset{\text{def}}{=} O_X \underset{f^\bullet O_Y}{\otimes} f^\bullet \mathcal{D}_Y \tag{24}$$

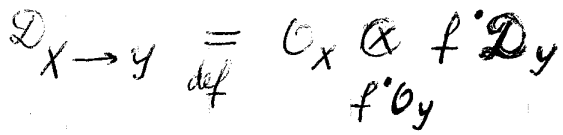
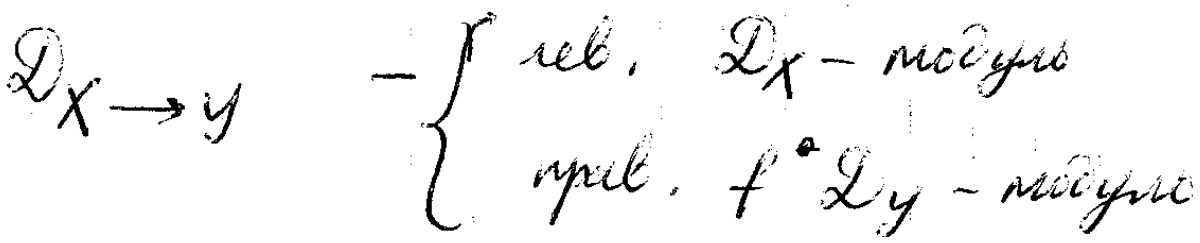


Diagram and typed description. In (24) the marks above both occurrences of f are filled dots.

Original source details: GELFAND-P000936

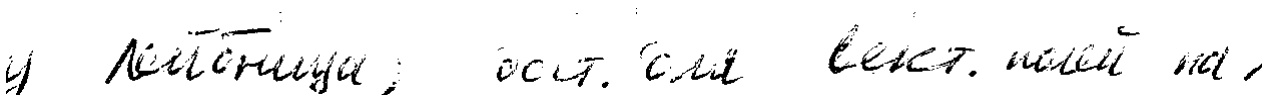
$$\mathcal{D}_{X\to Y} - \begin{cases} \text{left } \mathcal{D}_X\text{-module} \\ \text{right } f^\circ\mathcal{D}_Y\text{-module} \end{cases} \tag{25}$$



Original source details: GELFAND-P000936

Diagram and typed description. In the lower line of (25) the mark above f has an open center.

left $\mathcal{D}_X$ by the Leibniz rule; rem. for vec. fields on X:

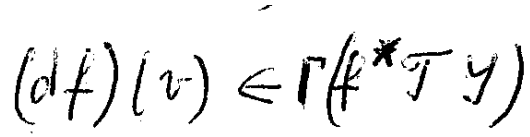


Original source details: GELFAND-P000936

Diagram and typed description. After the semicolon the manuscript has the abbreviation «ост.» (“rem.”).

$v \in$ Vect X;

$$(df)(v) \in \Gamma(f^* T\, Y)$$



Original source details: GELFAND-P000936

Diagram and typed description. After f^*, T and Y are written on the same line.

$(df)(v)$ acts on $\underline{\underline{f^\bullet \mathcal{D}_Y}}$.



Original source details: GELFAND-P000936

Uncertain reading. A heavy slanted stroke stands to the left of the last line. There is no visible separate dot below it.



Original source details: GELFAND-P000936

Diagram and typed description. A filled dot stands above f. The expression is double underlined.

GELFAND-P000937

17.11.80 Gelfand Sem.

I. M. G. problems.

1.

$$f(x) = \begin{cases} 1/q, & x = p/q \text{ — in lowest terms} \\ 0, & x \text{ — irrat.} \end{cases}$$

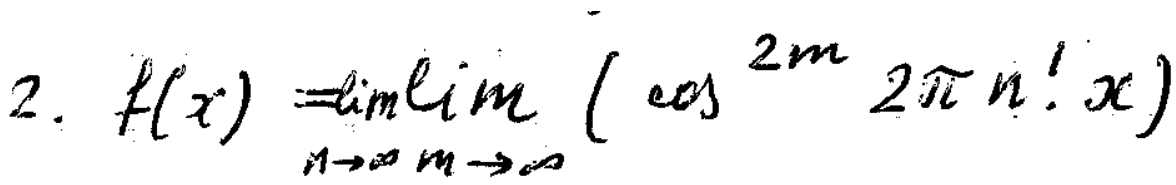
Where is it continuous?

where is it differentiable?

$$f(x) = \begin{cases} 1/q^\alpha, & x = p/q \text{ — in lowest terms} \\ 0, & x \text{ — irrat.} \end{cases}$$

What can be said about diff.?

2. $f(x) = \lim\limits_{n\to\infty} \lim\limits_{m\to\infty} (\cos^{2m} 2\pi n! x)$ — the Dirichlet fn.



Original source details: GELFAND-P000937

Diagram and typed description. The n in the cosine argument is followed by a factorial sign. The inner limit is over m.

3. Can the Dirichlet fn. be constructed with one limit sign?

(from cont. fns.)

[at upper right]

4. Calculate approximately

$$\int_{-\infty}^{\infty} e^{-x^2} \cos 1000x \, dx$$

The same with e^{-x^4} or e^{-x^6}

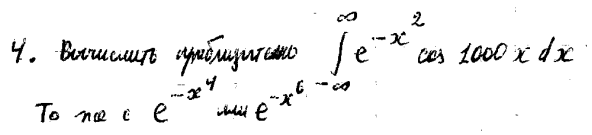


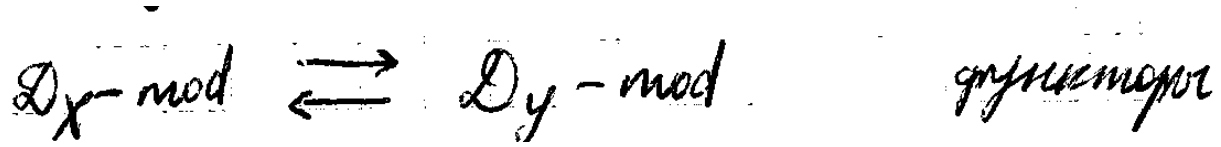
Diagram and typed description. Problem 4 is written in the upper right. The integral has $1000x$.

Original source details: GELFAND-P000937

A. Beilinson.

$X \xrightarrow{f} Y$, X, Y — varieties.

$\mathcal{D}_X$-mod, $\mathcal{D}_Y$-mod — categories of quasi-coher. sheaves of $\mathcal{D}_X$-mod and $\mathcal{D}_Y$-mod on X and Y



Original source details: GELFAND-P000937

Diagram and typed description. The upper arrow points right and the lower arrow points left. The arrows carry no labels.

$\mathcal{D}_X$-mod $\rightleftarrows$ $\mathcal{D}_Y$-mod

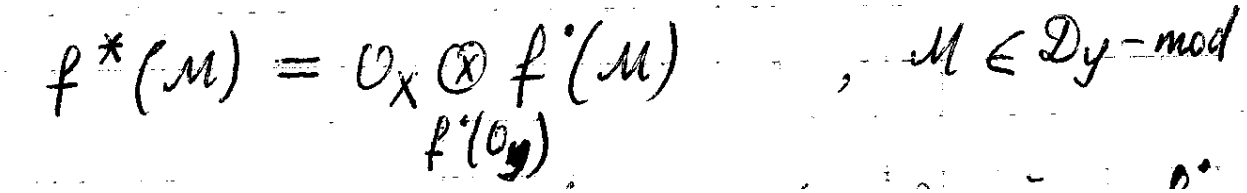
functors

Let us define:

$$f^* : \mathcal{D}_Y\text{-mod} \longrightarrow \mathcal{D}_X\text{-mod}$$

$$f^*(\mathcal{M}) = O_X \underset{f^\bullet(O_Y)}{\otimes} f^\bullet(\mathcal{M}), \qquad \mathcal{M} \in \mathcal{D}_Y\text{-mod} \tag{1}$$

(as a sheaf — inverse image in the categ. of O-modules, $f^\bullet$ — the abstract inverse image)



Original source details: GELFAND-P000937

Diagram and typed description. Both occurrences of f on the right of (1) have filled dots. The star is on the left.

Action of diff. operators: vec. fields suffice $v \in \mathrm{Vect}\, X$,

$$\ell \otimes m \in f^*(\mathcal{M}) \tag{2}$$

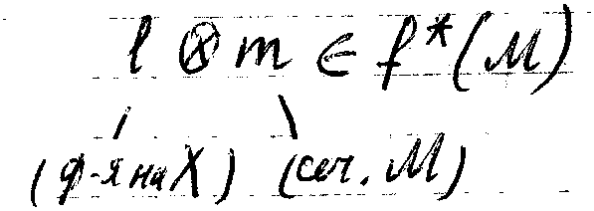


Diagram and typed description. Two leaders appear below the factors in (2).

$\ell \otimes m \in f^*(\mathcal{M})$

(fn. on X) (sect. of $\mathcal{M}$)

Original source details: GELFAND-P000937

$$df(v) = \sum_{\substack{\tau_i \\ | \\ \text{fn. on } X}} \otimes \frac{\partial}{\partial y_i} \in f^*(T_Y), \qquad df : T_X \longrightarrow f^* T_Y \tag{3}$$

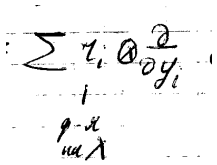


Diagram and typed description. The note under τ_i reads “fn. on X” in the singular.

Original source details: GELFAND-P000937

Instead of (1), one can write more simply

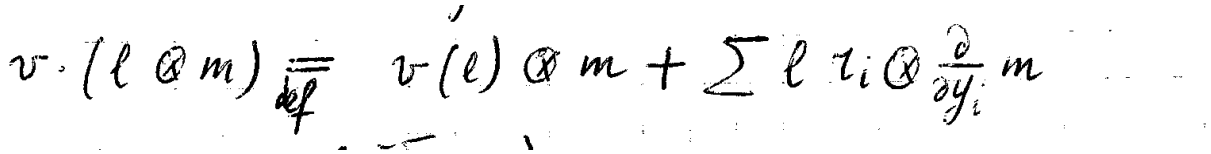
$$f^*(\mathcal{M}) = O_X \underset{O_Y}{\otimes} \mathcal{M} \tag{4}$$

In this same sense:

$$f^*(T_Y) = O_X \underset{O_Y}{\otimes} T_Y \tag{5}$$

$$v \cdot (\ell \otimes m) \underset{\text{def}}{=} v(\ell) \otimes m + \sum \ell \tau_i \otimes \frac{\partial}{\partial y_i} m \tag{6}$$

(by the Leibniz formula).



Original source details: GELFAND-P000937

Diagram and typed description. The equality in (6) has def below it. The coefficient in the sum reads $\ell\tau_i$.

GELFAND-P000938

[source number: 24]

If $\mathcal{M}$ comes from a vec. bundle with an integr. [inserted below: (flat)] connection, then this is the usual inverse image of the connection.

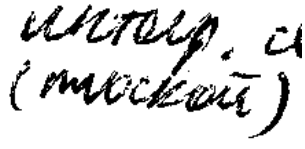


Diagram and typed description. “(Flat)” is inserted beneath “integr.”.

Original source details: GELFAND-P000938

(7) Examples.

1. $f^*O_Y = O_X$

2. From a sheaf [crossed out: unresolved sign] of fin. type there may result [inserted above: a sheaf] of infinite type



Undeciphered passage. The sign after “sheaf” is heavily crossed out. It may be $\mathcal{D}$.

Original source details: GELFAND-P000938

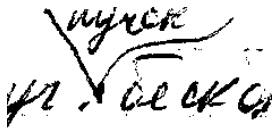


Diagram and typed description. “Sheaf” is inserted above the line with a leader.

Original source details: GELFAND-P000938

X — a point, Y — a var..

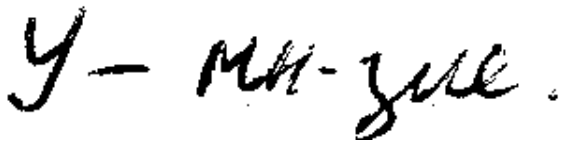


Diagram and typed description. The abbreviation after Y reads «мн-зие» (“var.”).

Original source details: GELFAND-P000938

$f^*(\mathcal{D}_Y)$ is infinite-dimensional (as a vec. space)

$$\mathcal{D}_X = \mathbb{C}$$

$f^*(\mathcal{D}_Y)$ — diff. op. with const. coefficients

3. X, Y as above, $\mathcal{M}$ — δ-fns. at the point $X \in Y \underset{\text{def}}{=} \delta_x$

(generalized fns. concentr. at a point).

Then

$$f^*(\delta_x) = 0,$$

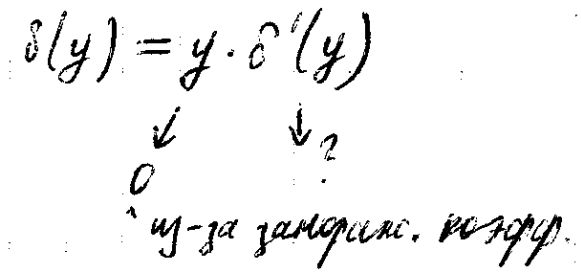
since $\forall$ generalized fn. [inserted above: at a pt.] can be repres. in the form

$$\sum \alpha_i s_i, \qquad \alpha_i(X) = 0.$$



Diagram and typed description. “At a pt.” is inserted above the line with a leader.

Original source details: GELFAND-P000938

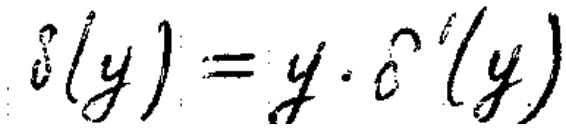


Original source details: GELFAND-P000938

Diagram and typed description. An arrow leads from y to 0, and one from $\delta'(y)$ to "?".

$$\delta(y) = \underset{\uparrow}{0} \cdot \underset{?}{\delta'(y)}$$

due to freez... coeff.



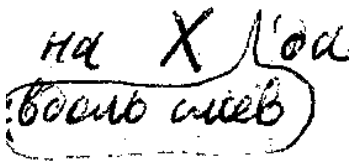
Original source details: GELFAND-P000938

Diagram and typed description. There is an equals sign with no visible minus before y.

Direct image is def. for right mod. (integr. forms only!, and they are a right mod.)

$$f_* \quad \text{mod-}\mathcal{D}_X \longrightarrow \text{mod-}\mathcal{D}_Y$$

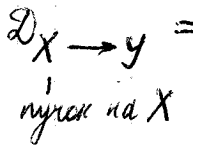
(integr. the volume forms on X [inserted below and circled: along the fibers] gives a volume form on Y)



Original source details: GELFAND-P000938

Diagram and typed description. The insertion "along the fibers" is circled below the line.

$$\underset{\text{a sheaf on } X}{\mathcal{D}_{X\to Y}} = f^*(\mathcal{D}_Y) \in \mathcal{D}_X\text{-mod-}\mathcal{D}_Y \tag{8}$$



Original source details: GELFAND-P000938

Diagram and typed description. A vertical leader extends down from the sheaf symbol in (8).

(a left $\mathcal{D}_X$-module and a right $\mathcal{D}_Y$-module)

$$f_*(\mathcal{M}) = f.\left(\mathcal{M} \underset{\mathcal{D}_X}{\otimes} \mathcal{D}_{X\to Y}\right) \tag{9}$$

$f.$ — direct image on sheaves

[crossed out: earlier number unresolved] (10) Examples.



Original source details: GELFAND-P000938

Uncertain reading. An earlier number is crossed out before (10). Its digit is unresolved.

1. $X = x \in Y$. Consid. the stand. 1-dim. $\mathcal{D}_X$-module $(= \mathbb{C})$



Diagram and typed description. The abbreviation before "1-dim." reads "stand.".

Original source details: GELFAND-P000938

$f_*(\mathcal{M})$ — (generalized fns. concentr. at the point on Y) $= \delta_x$

$$f^*(\mathcal{D}_Y) = \mathcal{D}_Y / \sum y_i D_Y$$

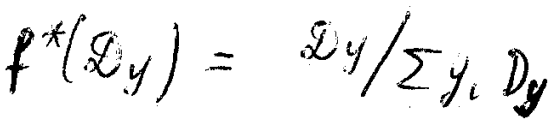
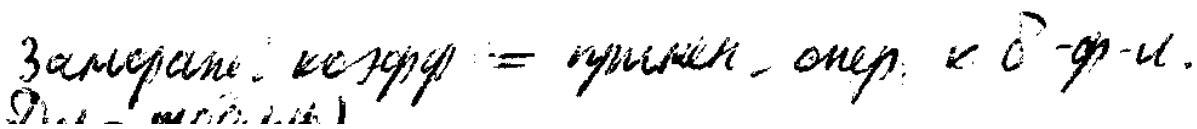


Diagram and typed description. The D in the denominator appears plain, without the calligraphic flourish of the numerator.

Original source details: GELFAND-P000938

[at right] Freez... (?) coeff. $=$ applying an op. to the δ-fn..



Original source details: GELFAND-P000938

Uncertain reading. The start of the word is tentatively read as "freez...". The exact ending is unresolved.

$(\Rightarrow \mathcal{D}_Y$-module)

GELFAND-P000939

2. Let us generalize:

$$X \xrightarrow{f} Y \qquad \text{embedding}$$

$f_*(\mathcal{M})$ by defn. — extension by means of δ-fns. in the transv. direction.

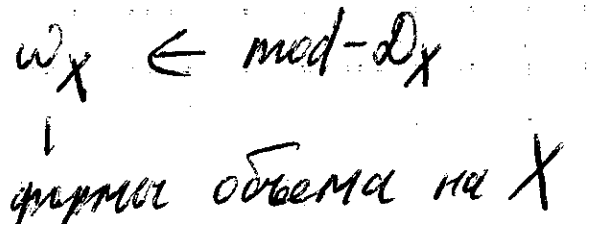


Diagram and typed description. A short stroke beneath ω_X leads to the caption.

$\omega_X \in$ mod-$\mathcal{D}_X$
|
volume forms on X

Original source details: GELFAND-P000939

$f_*(\omega_X)$ — generalized fns. concentr. on X and smooth in the dir. of X.

3. more generally:

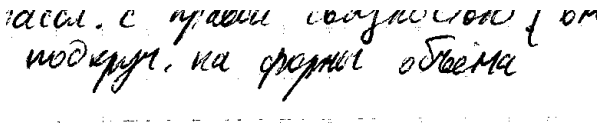
[crossed out: earlier number unresolved] Vector bundle with a right connection (in place of volume forms)



Uncertain reading. The crossed-out digit resembles 4. The exact reading is unresolved.

Original source details: GELFAND-P000939

[added below: twist. by volume forms]



Original source details: GELFAND-P000939

Diagram and typed description. “Twist. by volume forms” is added beneath the line about the connection.

Relation to cohomology.

[crossed out: earlier number unresolved] Y — a point, X — a var.



Original source details: GELFAND-P000939

Uncertain reading. The earlier number before “Y is a point” is crossed out. It may be 4.

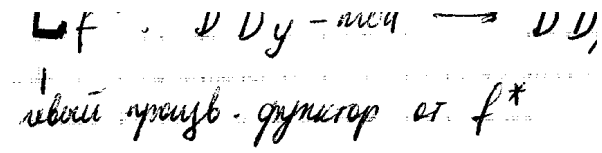
$f^*(\mathcal{M})$ — the sheaf of fns., rep. in the requisite number of copies.

f_* here computes the cohomology of the space X.

Higher direct images.

$$\begin{array}{c} Lf^* : D\,D_Y\text{-mod} \longrightarrow D\,D_X\text{-mod} \\ | \end{array} \tag{11}$$

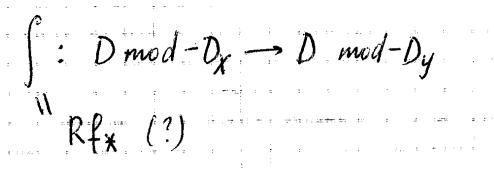
left derived functor of f^*.



Original source details: GELFAND-P000939

Diagram and typed description. A vertical stroke below L leads to the caption.

D — derived category

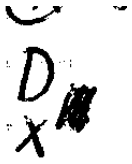


Original source details: GELFAND-P000939

Diagram and typed description. Two slanted strokes beneath the integral sign lead to “Rf_* (?)”.

$$\begin{array}{l} \int : D\,\text{mod-}D_X \\ \quad \backslash\backslash \quad \longrightarrow D\,\text{mod-}D_Y \\ \quad Rf_*\,(?) \end{array} \tag{12}$$

$$\int \mathcal{M} = Rf_\bullet \left(\mathcal{M} \overset{L}{\underset{D_X}{\otimes}} \mathcal{D}_{X \to Y} \right) \quad \text{— a complex} \tag{13}$$



Original source details: GELFAND-P000939

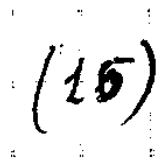
Uncertain reading. An X appears below D. The earlier sign crossed out beside it is unresolved.

$$\int^i \mathcal{M} = H^i\left(\int \mathcal{M}\right) \tag{14}$$

(Sato, Kashiwara)

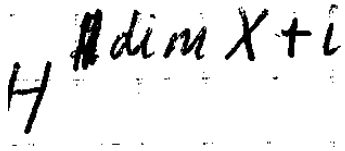
(15) Theorem. Y — a point $\Rightarrow$

$$\int^i \omega_X = H^{\dim X+i}(X) \tag{16}$$



Original source details: GELFAND-P000939

Diagram and typed description. The formula is numbered (16). The lower loop of the last digit is closed.



Original source details: GELFAND-P000939

Undeciphered passage. A sign is crossed out before dim $X + i$ in the exponent. The exact reading is unresolved.

(X — nonsingular over $\mathbb{C}$)

GELFAND-P000940

[source number: 25]

S. Gelfand

Derived category.

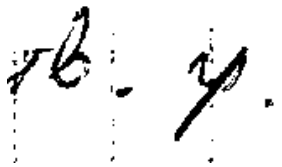


Original source details: GELFAND-P000940

Diagram and typed description. The heading has “category” in the singular.

Prob. 1. If there are 2 compl. with isom. cohomologies, one cannot always extend an isom. of cohom. to a map of compl. inducing an isomorphism.

Prob. 2. Consid. the sphere S^n and on it the cat. of sheaves of ab. groups C. Acting on it is $O(n)$. Consid. the cat. with the action of the grp. and the derived cat. $D(C)$. Consid. the dual S^{*n}. C_1 [crossed out: unresolved]



Original source details: GELFAND-P000940

Diagram and typed description. After “action” the abbreviation reads «гр.» (“grp.”).



Original source details: GELFAND-P000940

Undeciphered passage. A sign to the right of the first C_1 is heavily crossed out.

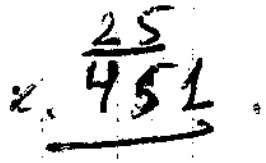
C_1 is not isom. to C (the action is different)

$$\underline{\underline{D(S) \sim D(S^*)}} \quad !!$$

24.11.80 Gelfand Sem.

Sectional sessions.

25 rm. $\overrightarrow{451}$ [crossed out: unresolved]



Original source details: GELFAND-P000940

Uncertain reading. The abbreviation before 451 is tentatively read as «к.» (“rm.”).



Original source details: GELFAND-P000940

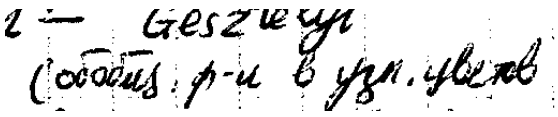
Undeciphered passage. The crossed-out writing before 15^{00} is unresolved.

$15^{\underline{00}}$ Sergeev
$15^{\underline{20}}$ Mebkhout (France)
Riem.-Hilb. prob. in higher dim.
$15^{\underline{40}}$ Henkin
$16^{\underline{00}}$ Kantor J.-M.

[boxed at right]

Conf. Plenary sessions.

24. $16^{\underline{20}}$ Sato
25 $10^{\underline{00}}$ Morimoto
$10^{\underline{40}}$ Maslov
$11^{\underline{40}}$ Gesztelyi
(generalized fns. in eqs. partial (?)
$12^{\underline{20}}$ Berezanskii et al.



Original source details: GELFAND-P000940

Uncertain reading. The ending after «в ур.» is tentatively read as «частн.» (“partial”). No closing parenthesis is visible.

26. [crossed out: unresolved] 10.00 Laugwitz
(from a nonstand. point of view)
$10^{\underline{40}}$ Ivanov
$11^{\underline{40}}$ Khromov, Danilov
$12^{\underline{20}}$ Berg multiplic. of distrib.



Original source details: GELFAND-P000940

Undeciphered passage. The earlier time before 10.00 is heavily crossed out.

28. $10^{\underline{00}}$ Lions
$10^{\underline{40}}$ Gelfand
$11^{\underline{40}}$ Komatsu — kernels in ultradistrib.
$12^{\underline{20}}$ Lassner
$13^{\underline{00}}$ Zharinov

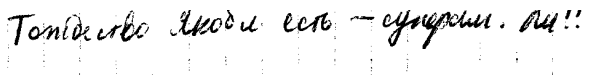
D. B. Fuchs. Whitehead mult.

$$\pi_n(X) \times \pi_m(X) \longrightarrow \pi_{n+m}(X)$$

$\sim$ commutator in a Lie alg.

$$\partial(I_n \times I_m) = (\partial I_n \times I_m) \cup (I_n \times \partial I_m)$$

There is a Jacobi identity — Lie supersym.!!



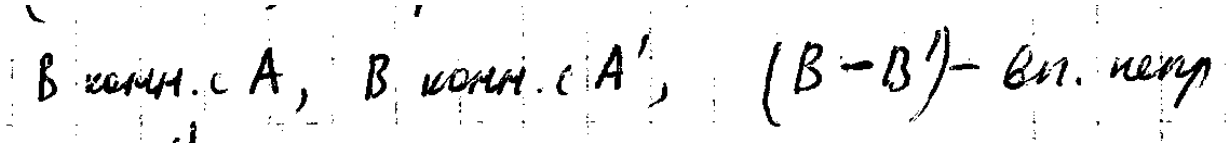
Original source details: GELFAND-P000940

Diagram and typed description. The abbreviation before “Lie” reads «суперсим.» (“supersym.”).

V. Lomonosov.

$(A - A')$ — nuclear

B comm. with A, B comm. with A', $(B - B')$ — compl. cont.



Original source details: GELFAND-P000940

Diagram and typed description. The B before A' has no visible prime.

$$\Rightarrow \quad B|_{\mathcal{H}_{ac}(B)} \underset{\text{unit.}}{\sim} B'|_{\mathcal{H}_{ac}(B')}$$

$\Rightarrow$ the theor. is consid. directly for commut. families of operators.

A. Beilinson.

$X \xrightarrow{f} Y$

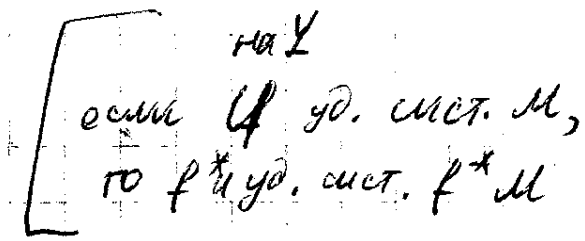
$$f^* : \mathcal{D}_Y\text{-mod} \longrightarrow \mathcal{D}_X\text{-mod} \tag{1}$$

Introduce

$$f_* : \text{mod-}\mathcal{D}_X \longrightarrow \text{mod-}\mathcal{D}_Y$$

[at right]

$$\left[\begin{array}{l} \text{if } \overset{\text{on } Y}{u} \text{ sat. the syst. } M, \\ \text{then } f^*u \text{ sat. the syst. } f^*M \end{array}\right.$$



Original source details: GELFAND-P000940

Diagram and typed description. “On Y” is inserted above u in the upper note on the right.

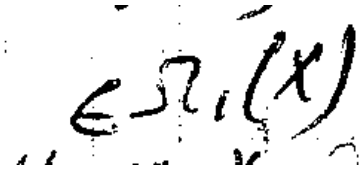
$$\mathcal{D}_{X\to Y} = f^*(\mathcal{D}_Y) \in \mathcal{D}_X\text{-mod-}\mathcal{D}_Y \qquad \text{(bimodule)} \tag{2}$$

$$f_*M = f_\bullet\left(M \underset{\mathcal{D}_X}{\otimes} \mathcal{D}_{X\to Y}\right) \tag{3}$$

(f_* on volume forms of comp. support)
if

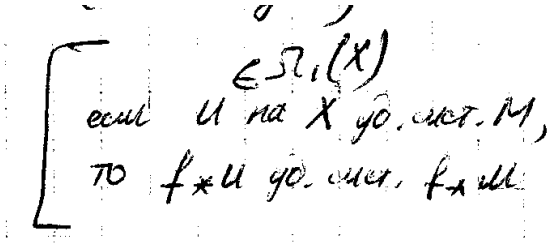
[at right]

$$\left[\begin{array}{l}\text{if } u \overset{\in\Omega_1(X)}{\text{ on }} X \text{ sat. the syst. } M,\\ \text{then } f_*u \text{ sat. the syst. } f_*M\end{array}\right.$$



Original source details: GELFAND-P000940

Uncertain reading. The small subscript on Ω resembles 1. The exact reading is unresolved.



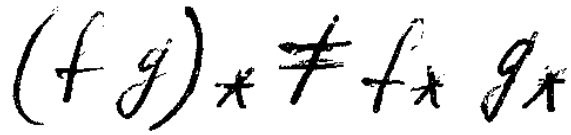
Original source details: GELFAND-P000940

Diagram and typed description. The lower note on the right begins with an opening square bracket.

GELFAND-P000941

(System of eqs. satisfied by improper integrals)

(4) f_* is not functorial: $(fg)_* \neq f_*g_*$



Original source details: GELFAND-P000941

Diagram and typed description. In (4) fg and f_*g_* are written in this order.

Let us construct.

$$Rf_* : D(\text{mod-}D_X) \longrightarrow D(\text{mod-}D_Y) \tag{5}$$

Example: X — a plane, Y — a line, f — projection,
$\mathcal{M}$ — a sheaf of forms on X (of top degree)

$Rf_*(\mathcal{M})$ — a complex, $\quad$ 0-comp. — 0
(-1)-comp. — forms on the line.



Original source details: GELFAND-P000941

Diagram and typed description. The example gives the 0-comp. and the (-1)-comp.

$$Rf_*(\mathcal{M}) = Rf_\bullet\left(\mathcal{M} \overset{L}{\underset{\mathcal{D}_X}{\otimes}} \mathcal{D}_{X\to Y}\right) \tag{6}$$

$\overset{L}{\otimes}$ — the left derived functor of the tens. product.

$Rf_\bullet$ — the right derived functor of the functor (!) $f_\bullet$

(7) Prop.

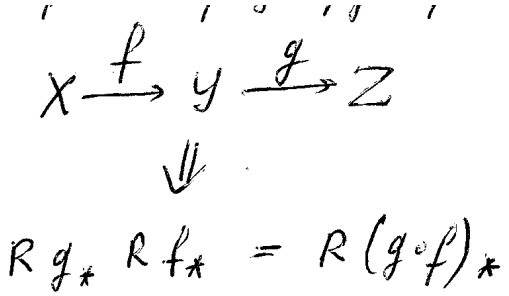


Diagram and typed description. A double arrow below the chain $X \to Y \to Z$ points down to the formula.

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$
$$\Downarrow$$
$$Rg_*Rf_* = R(g\circ f)_*$$

Original source details: GELFAND-P000941

[crossed out: ~~since~~] (f_* lacks this prop.)



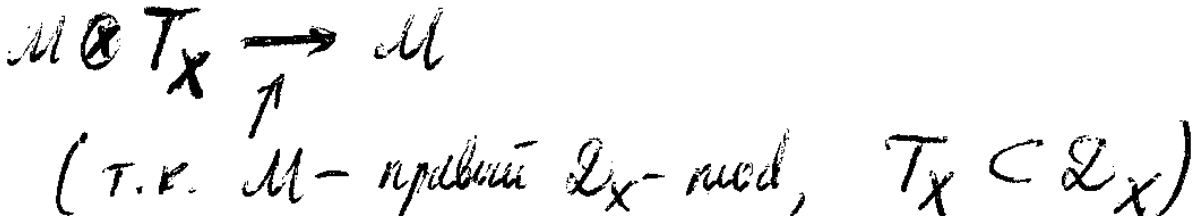
Uncertain reading. An opening parenthesis and the start of “since” are crossed out.

Original source details: GELFAND-P000941

Let

$$\mathcal{M} \in \text{mod-}\mathcal{D}_X$$
$$\ldots \longrightarrow \mathcal{M}\otimes\Lambda^2 T_X \longrightarrow \mathcal{M}\otimes T_X \underset{\nearrow}{\longrightarrow} \mathcal{M} \tag{8}$$

(T_X — tangent sheaf to X) (since $\mathcal{M}$ — a right $\mathcal{D}_X$-mod, $T_X \subset \mathcal{D}_X$)



Original source details: GELFAND-P000941

Diagram and typed description. An arrow from the right-hand note points toward the last differential in (8).

(9) Prop.

Consid. the cohomology of (8). It agrees with the cohom. of the compl. (6), i.e.

(de Rham!)

$$R\Gamma(\underbrace{DR}_{\text{De Rham}}\ \mathcal{M}) = Rf_*(\mathcal{M}) \tag{10}$$

(here $f : X \longrightarrow$ a point).

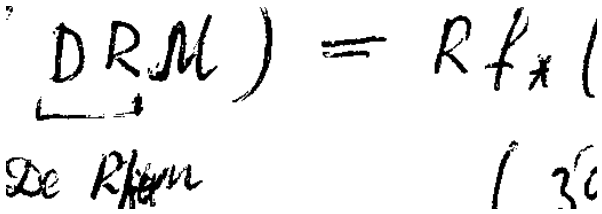


Diagram and typed description. Below DR a bracket leads to the label “De Rham”.

Original source details: GELFAND-P000941

$\mathcal{M}$ — the sheaf of volume forms $\Rightarrow$ (8) — the usual de Rham compl., shifted by $\mathcal{M}$ to the left



Uncertain reading. The sign after “by” resembles the $\mathcal{M}$ at the beginning of the sentence.

Original source details: GELFAND-P000941

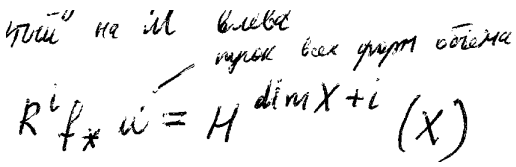


Diagram and typed description. A slanted stroke leads from the annotation toward ω.

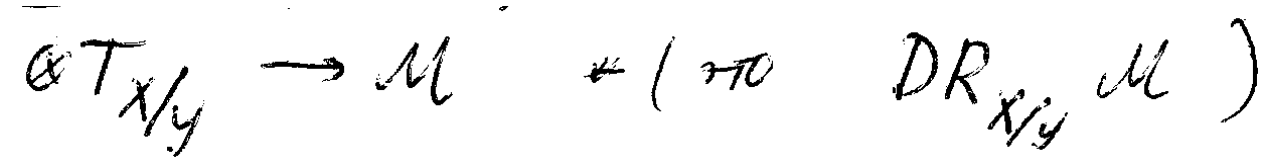
sheaf of all volume forms

$$R^i f_*\ \omega = H^{\dim X + i}(X) \tag{11}$$

Original source details: GELFAND-P000941

More gen.: For a smooth morphism, sheaves of cohom. along the fibers!

$$\ldots \longrightarrow \mathcal{M} \otimes \Lambda^2 T_{X/Y} \longrightarrow \mathcal{M} \otimes T_{X/Y} \longrightarrow \mathcal{M} \quad \leftarrow \quad (\text{this is } DR_{X/Y}\mathcal{M}) \tag{12}$$



Original source details: GELFAND-P000941

Diagram and typed description. The short arrow points left from the parenthetical note toward the complex.

GELFAND-P000942

[source number: 26]

Then

$$Rf_\bullet(DR_{X/Y}\mathcal{M}) = Rf_*(\mathcal{M}) \tag{13}$$

Thus in the direct image of [inserted above: rel.] DeRhama one introduces the action of diff. operators (!)

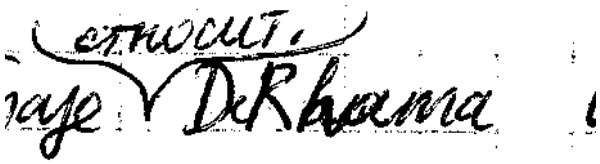
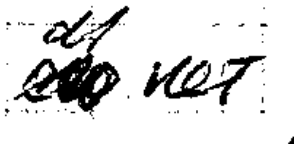


Diagram and typed description. The line reads “DeRhama”. The abbreviation for “relative” is inserted above its start.

Original source details: GELFAND-P000942

(this is not obvious, since in the orig. sheaf [crossed out: ~~it~~ (?)] [inserted above: unresolved] is absent). This action is called



Uncertain reading. The lower word before “absent” is heavily retraced and tentatively reads “it”. A short insertion above it is unresolved.

Original source details: GELFAND-P000942

the Gauss–Manin connection.

2nd case

$X \xrightarrow{f} Y$ embedding.

$$\begin{aligned} Rf_*(\mathcal{M}) &= Rf_\bullet\left(\mathcal{M} \overset{L}{\underset{\mathcal{D}_X}{\otimes}} \mathcal{D}_{X\to Y}\right) \\ &= f_\bullet\left(\mathcal{M} \overset{L}{\underset{\mathcal{D}_X}{\otimes}} \mathcal{D}_{X\to Y}\right) = \end{aligned} \tag{14}$$

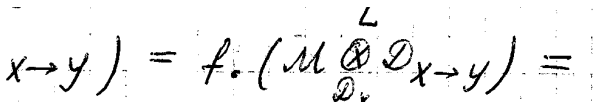


Diagram and typed description. At the end of the manuscript line (14) another equals sign follows the second expression.

Original source details: GELFAND-P000942

(The functor $f_\bullet$ is exact in this case —)

— no cohom. arises along the fibers

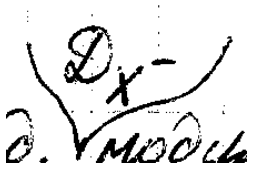
$$= f_\bullet \left(\mathcal{M} \underset{\mathcal{D}_X}{\otimes} \mathcal{D}_{X \to Y} \right)$$

(since $\otimes$ is exact in this case)

$\omega \rightsquigarrow$ forms extended by δ-fns. in the transv. direction.

Holonomic modules.

X — a var., $\mathcal{M}$ — a finitely generated $\mathcal{D}_X$-module [crossed out: unresolved]



Original source details: GELFAND-P000942

Diagram and typed description. $\mathcal{D}_X$- is inserted above "module".



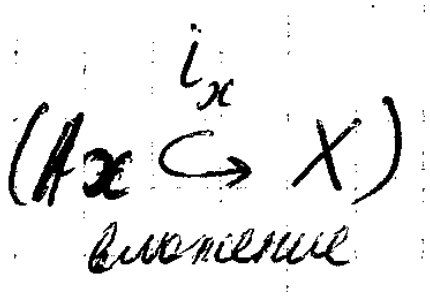
Original source details: GELFAND-P000942

Undeciphered passage. The writing after "module" is crossed out. It resembles $\mathcal{M}$, but the reading is unresolved.

It is called holonomic, if $\forall\, x \in X$,

$$\text{the cohomology of} \quad Li_x^* \mathcal{M} \quad \text{is finite-dimensional}$$

(stalk of the module over the point)



Original source details: GELFAND-P000942

Diagram and typed description. The arrow i_x goes from x to X. A heavy unresolved stroke precedes x.

$$([?]\, x \xrightarrow{i_x} X)$$

embedding

($i_x^* \mathcal{M}$ — "freezing coeff.", see above)

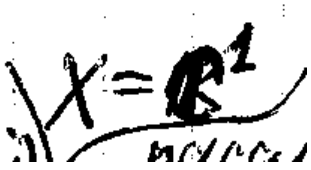
$\sim$ the complex $Li_x^* \mathcal{M}$ is finite-dimensional (since in the derived category).

A free module is not holonomic (see last time)

Holonomicity $\sim$ overdeterminacy.

Examples

a) [inserted above: $X = \mathbb{R}^1$ (?)] a bundle with an integr. connection



Original source details: GELFAND-P000942

Diagram and typed description. The field letter is retraced and has a slanting lower stroke. The tentative reading is $\mathbb{R}^1$.

The module is of this form except at a finite number of points where there may be sing.

b) $X = \mathbb{C}^2$ again on the complement of finitely many [crossed out: ~~points~~ (?)] [inserted above: curves] — a bundle with an integr. conn.



Original source details: GELFAND-P000942

Diagram and typed description. An earlier word is crossed out below the inserted “curves”. It may read “points”.

There are singularities on these curves.

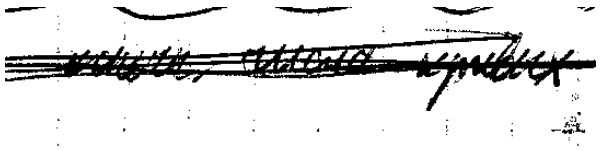
(?) All are obtained by gluing from modules obtained from δ-fns. in the normal direction.



Original source details: GELFAND-P000942

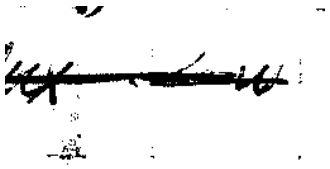
Diagram and typed description. A circled question mark stands to the left of the line about gluing.

[crossed out: ~~(except for points [...] curves~~ unresolved]



Original source details: GELFAND-P000942

Uncertain reading. “Points” is tentatively read in the middle of the crossed-out line. The next word is unresolved.



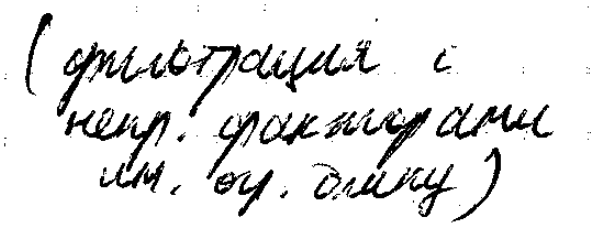
Original source details: GELFAND-P000942

Undeciphered passage. There is further crossed-out writing after “curves”. The ending is unresolved.

GELFAND-P000943

(15) Thm. 1) $\forall$ holon. module has finite length

(filtrations with irred. factors have the same length)



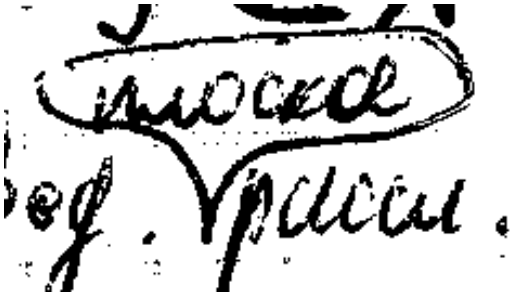
Original source details: GELFAND-P000943

Uncertain reading. The ending of the first word in parentheses is indistinct. The last line reads “have the same length”.

2) there is a descr. of all irreducibles

$$Y \subset X, \qquad Y \text{ irred.}$$

Irred. [inserted above and circled: flat] bundle $\mathcal{L}_Y$ with a connection on Y
(a vector bundle with an integr. conn.)



Original source details: GELFAND-P000943

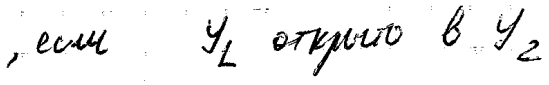
Diagram and typed description. A leader points from the circled “flat” to “bundle”.

flat

Irred. bundle

2 pairs Y_1, Y_2 are called equiv. if Y_1 is open in Y_2
and

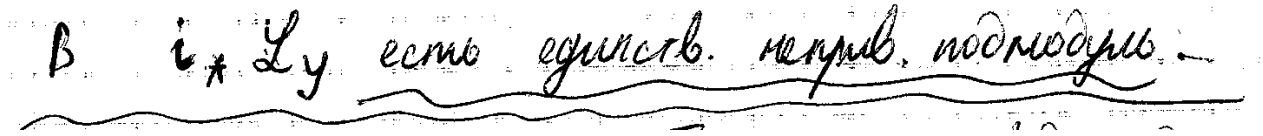
$$\mathcal{L}_{Y_1} = (\mathcal{L}_{Y_2})|_{Y_1}$$



Original source details: GELFAND-P000943

Diagram and typed description. The condition is written in the order “Y_1 is open in Y_2”.

(one can discard closed subsets, spec. over the generic point)
(16) Assertion. $i : Y \hookrightarrow X$
In $i_*\mathcal{L}_Y$ there is a unique irred. submodule —



Original source details: GELFAND-P000943

Diagram and typed description. Two wavy underlines of different lengths appear below the line.

this is the object sought, an irred. module
Y — smooth $\Rightarrow$ already irreducible
At sing. there appear δ-fns..



Original source details: GELFAND-P000943

Diagram and typed description. Below the last line are two tapered strokes with a dot between them.



GELFAND-P000944

[source number: 27]

1/XII-80 Gelfand Sem.

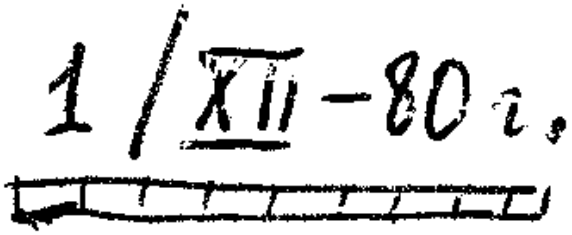


Diagram and typed description. A narrow bar with transverse strokes underlines the date. Separate lines appear above and below XII.

1/XII-80

Original source details: GELFAND-P000944

S. P. Novikov.

$$\hat{H} = \frac{1}{2}\sum_{\alpha=1}^{n}\left(i\frac{\partial}{\partial x^{\alpha}} - eA_{\alpha}\right)^{2} + cV(\vec{x}) \tag{1}$$



Undeciphered passage. Heavily crossed-out writing appears after the equals sign in (1), before the fraction. Its reading is unresolved.

Original source details: GELFAND-P000944

$$H_{\alpha\beta} = \partial_{\beta}A_{\alpha} - \partial_{\alpha}A_{\beta} \tag{2}$$

V — el. field, $H_{\alpha\beta}$ — magn. field

$$\hat{H}\psi = \varepsilon\psi \qquad \text{spectr. problem.} \tag{3}$$

The interesting case is when

(4) $H_{\alpha\beta}(x)$ and $V(x)$ are periodic with a lattice of periods.

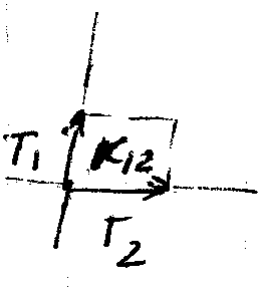
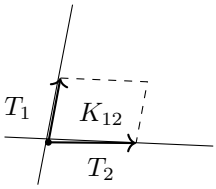
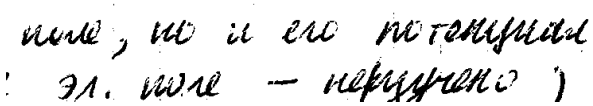


Diagram and typed description. Cell K_{12} lies between the arrows T_1, T_2. The top and right sides are broken lines.



Original source details: GELFAND-P000944

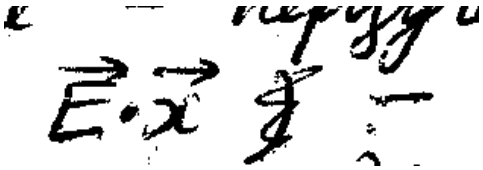
(not only the el. field but also its potential. A constant el. field — unreasonable)



Uncertain reading. The word after the dash is tentatively read as «неразумно» (“unreasonable”). Its ending is faint.

Original source details: GELFAND-P000944

— the potential then $\vec{E} \cdot \vec{x}$ [crossed out: unresolved] —



Original source details: GELFAND-P000944

Undeciphered passage. One sign is crossed out after $\vec{E} \cdot \vec{x}$. The exact reading is unresolved.

— even the 1-dim. problem is unstudied.)

A_α is nonperiodic (generally speaking).

$K_{\alpha\beta}$ — an elem. cell, spanned by the α and β coordinate



Original source details: GELFAND-P000944

Uncertain reading. The overwritten preposition after “spanned” is tentatively read as «на».



Original source details: GELFAND-P000944

Uncertain reading. The end of the word for “coordinate” is overwritten. The last letter reads «у».

$$\iint_{K_{\alpha\beta}} H_{\alpha\beta}\, dx^\alpha \wedge dx^\beta = \Phi_{\alpha\beta} \qquad \text{— magn. fluxes (basis)} \tag{5}$$

(no summation!)

$$i\nabla_\alpha = i\frac{\partial}{\partial x^\alpha} - eA_\alpha \qquad \text{has the meaning of a connection} \tag{6}$$

Rem. The spin here is 0, with spin $1/2$ one has

$$\hat{H} = (1) + e\sigma_j H^j, \qquad \text{where } H^1 = H_{23}, \quad H^2 = -H_{13}, \quad H^3 = H_{12}; \tag{7}$$

$\sigma_1, \sigma_2, \sigma_3$ — Pauli matr.

(8) Admissible case: $\frac{e}{2\pi}\Phi_{\alpha\beta} \in \mathbb{Z}$

The operator $\hat{H}$ is consid. in $L^2(\mathbb{R}^n)$.

(9) $(8) \Rightarrow \hat{H}$ — a univ. cover. op. on the torus (in a bundle), moreover



Original source details: GELFAND-P000944

Uncertain reading. The beginning of the word for “moreover” is retraced.

$$\left(c_1(\xi), \quad \underset{\substack{| \\ \text{basis cycle}}}{Z_{\alpha\beta}} \right) = \frac{e}{2\pi} \Phi_{\alpha\beta} \tag{10}$$

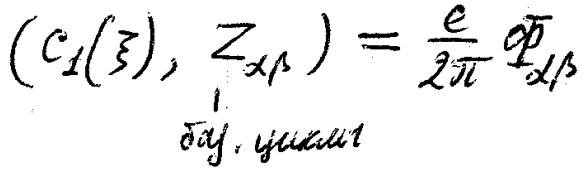
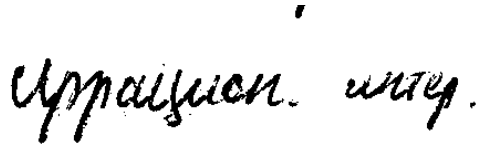


Diagram and typed description. A short stroke below $Z_{\alpha\beta}$ leads to “basis cycle”.

Original source details: GELFAND-P000944

But the op. always makes sense (phys.), yet there is no theory.

(11) Irrational is of int. Ration. reduces to it by coarsening the lattice



Uncertain reading. The abbreviation after “Irrational” is tentatively read as «интер.» (“of int.”).

Original source details: GELFAND-P000944

GELFAND-P000945

Magnetic translations.

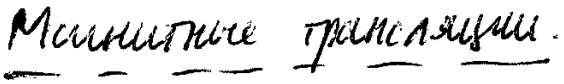
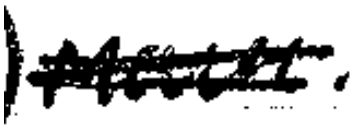


Diagram and typed description. A broken line underlines the heading.

Original source details: GELFAND-P000945

2 types of problems: a) [crossed out: ~~magn.~~ (?)] the lattice is imposed (solid state!), and the magn. field is external

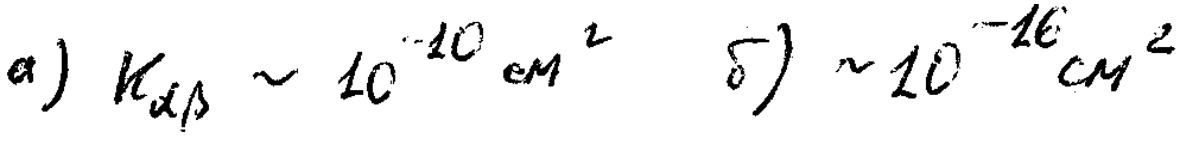


Uncertain reading. The crossed-out abbreviation before “lattice” is tentatively read as “magn.”.

Original source details: GELFAND-P000945

b) of superconduct. type: the magn. field arises on its own, they are automatically integral.

a) $K_{\alpha\beta} \sim 10^{-10}$ cm^2 b) $\sim 10^{-16}$ cm^2



Original source details: GELFAND-P000945

Diagram and typed description. The line reads 10^{-10} and 10^{-16} in square centimeters.

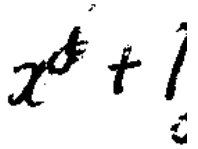
Magn. transl. We have:

$$A_\alpha(\dots, x^i + T_i, \dots) = A_\alpha(x) + \partial_\alpha f_i \tag{12}$$

(since dA is periodic)

$$\hat{T}_j^* \psi(x) = \psi(.. \; x^j + T_j, \;) \exp(-ie f_j) \tag{13}$$

(operator of magn. translation)

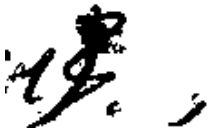


Uncertain reading. In (13) the superscript of x is retraced. The reading j is tentative and agrees with T_j on the same line.

Original source details: GELFAND-P000945

(14) Lemma 1 (known ~1960s [crossed out: unresolved], Zak and Brown)

$$\hat{T}_j^* \hat{T}_k^* = \hat{T}_k^* T_j^* \exp(-ie\Phi_{kj}), \qquad j, k = 1, \dots, n.$$



Uncertain reading. A sign after "1960s" is heavily retraced. Its exact reading is unresolved.

Original source details: GELFAND-P000945

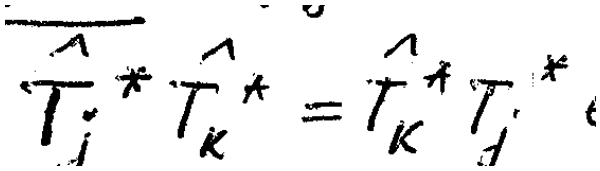


Diagram and typed description. The last T_j^* on the right side of the equality has no visible hat.

Original source details: GELFAND-P000945

This gives a projective repr. of an ab. group, with cocycles specified in advance.
Irrat. fluxes $\Rightarrow$ one gets groups not of type 1. (subgr. of the Heisenberg grp.).

$$\frac{e\Phi_{kj}}{2\pi} \in \mathbb{Z} \Rightarrow \text{commut. group} \tag{15}$$

$n = 2$
($n = 3$ is also interesting)
superconduct. is a 2-dim. problem
Problem: describe the spectrum of $\hat{H}$.
(16) $[\hat{H}, \hat{T}_j^*] = 0 \Rightarrow$ the eigenfns. must be sought as simultaneous eigenfns. for $\hat{T}_j^*$
[crossed out below the line: ~~magn.~~ (?)]



Uncertain reading. Below line (16) the crossed-out abbreviation resembles "magn.".

Original source details: GELFAND-P000945

Magnetic Bloch:

$$\hat{T}_j^* \psi_p = \exp(ip_j T_j)\psi_p \tag{17}$$



Uncertain reading. The left ψ subscript is heavily retraced. p is tentatively supplied from the neighboring line.

Original source details: GELFAND-P000945



Uncertain reading. The right ψ subscript is also heavily retraced. p is tentatively supplied.

Original source details: GELFAND-P000945

Op. in $L^2 =$ direct integral of op. on the torus with such bound. conds.

p_j — quasimomenta $\quad (p_1, p_2) \in \mathbb{T}^2 \quad$ (torus of the recip. lattice)

Fix $p_1, p_2 \Rightarrow$ an op. in a bundle on the torus with eigenvals.

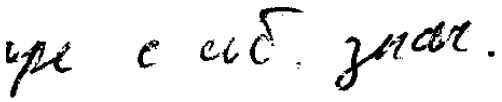


Diagram and typed description. The last two abbreviations read “eigenvals.”.

Original source details: GELFAND-P000945

$$\varepsilon_j(p_1, p_2) \qquad \text{— dispersion law} \tag{18}$$
$$j = 1, 2, \dots$$

GELFAND-P000946

[source number: 28]

(19) Theorem. Let $V = 0$, spin $s = \frac{1}{2}$,

$$n = 2, \qquad N \underset{\text{def}}{=} \frac{e\Phi_{12}}{2\pi}.$$

[In the right margin, beyond the double line]

Dubrovin, Novikov

ZhETF, Sept. 1980.

and there is also a paper in DAN

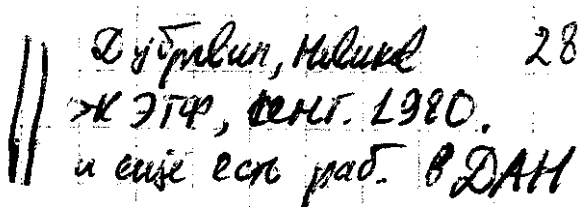
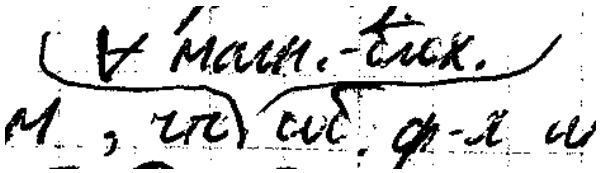


Diagram and typed description. References in the right margin.

Original source details: GELFAND-P000946

(magn. field directed along the Z axis — 2-dim. case — and periodic)

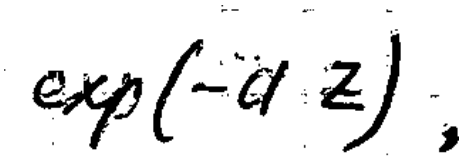
Then given $A = (a_1, \dots, a_N) \in C^N$, we get that the eigenfn. has the form [inserted above "eigen" ($\forall$ magn.-Bloch)]



Original source details: GELFAND-P000946

Diagram and typed description. A curved stroke links the insertion above the line to the main text.

$$\Psi_A = \exp(-e\varphi) \prod_{j=1}^{N} \sigma(z - a_j) \exp(-az). \tag{20}$$

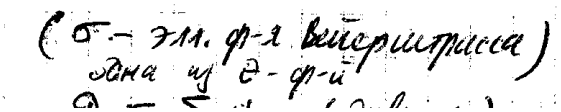


Original source details: GELFAND-P000946

Diagram and typed description. The sign before az in (20) is minus.

(σ — Weierstrass elliptic fn.)

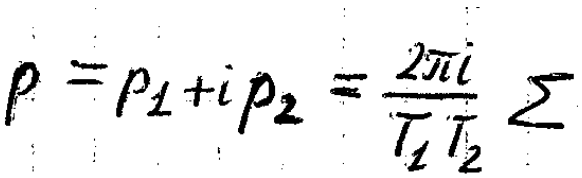
one of the θ-fns.



Original source details: GELFAND-P000946

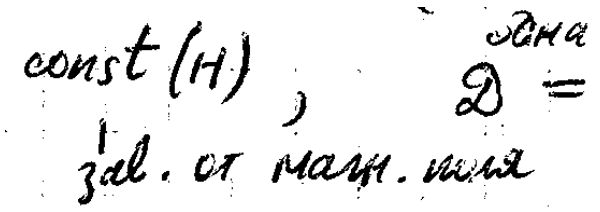
Diagram and typed description. Note at the right of (20).

$$p = p_1 + ip_2 = \frac{2\pi i}{T_1 T_2} \sum a_j + \underbrace{\text{const}(H)}_{\text{depends on the magn. field}}, \tag{21}$$



Original source details: GELFAND-P000946

Diagram and typed description. The summation term in (21) has a plus sign.



Original source details: GELFAND-P000946

Diagram and typed description. A line connects the annotation below const(H) to that term.

$$\mathcal{D} = \sum a_j. \qquad \text{(divisor)}$$

(22) [crossed out: ~~$\varphi(x,y)$~~ (?)]

$$\varphi(x,y) = \frac{1}{2\pi} \int_{K_{12}} \ln|\sigma(z-z')| H(z')\, d^2 z'.$$



Original source details: GELFAND-P000946

Uncertain reading. The crossed-out entry before (22) is tentatively read as $\varphi(x,y)$.

$z =$ [crossed out: entry unresolved] $x + iy$.



Original source details: GELFAND-P000946

Undeciphered passage. Several signs are crossed out after the equals sign. Their exact reading is unresolved.

Fns. (20) — a complete basis of solutions to the eq. $\hat{H}\psi = 0$

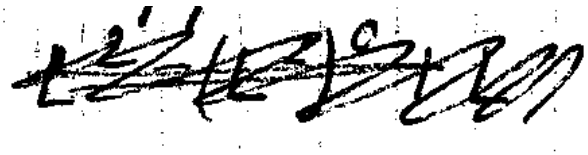
(exact integr. at the lower zero energy level)



Original source details: GELFAND-P000946

Diagram and typed description. The abbreviation after “exact” reads “integr.”

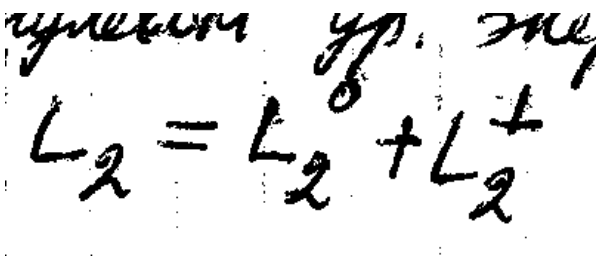
[crossed out: $L^2 = (L^2)^0 + \ldots$ (?)]



Original source details: GELFAND-P000946

Undeciphered passage. The beginning of the crossed-out formula is a tentative reading. The end after the plus sign is unresolved.

$$L_2 = L_2 + L_2^{\perp}.$$



Original source details: GELFAND-P000946

Diagram and typed description. The loop above the first L_2 on the right continues into a descender from the line above.

$$a = a(a_1, \dots, a_N) \quad \text{— is calculated.} \tag{23}$$

Selecting the dependence (23) — is selecting real quasimomenta

(the fn. Ψ_A is always an eigenfn. and Bloch).

$\lambda\Psi_A$ is also an eigenfn.

$(\lambda, a_1, \dots, a_N)$ — the full set of parameters.

Rem.

$$(\lambda, a_1, \dots, a_N) \xrightarrow{P} \mathbb{T}^2. \tag{24}$$

(25) We get [crossed out: entry unresolved] a holom. vect. bundle (N-dimensional) over $\mathbb{T}^2$ with $c_1 = N$.



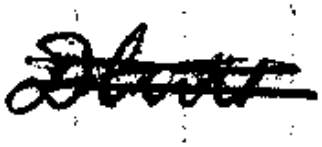
Original source details: GELFAND-P000946

Undeciphered passage. The heavily crossed-out entry before “holom.” is unresolved.

(= the number of quanta of magn. flux)

(26) Fibre — at fixed p — We get $\mathcal{L}(\mathcal{D})$, $\dim \mathcal{L}(\mathcal{D}) = N$

[crossed out: ~~Next~~ (?)] (mult. by a doubly periodic fn.)



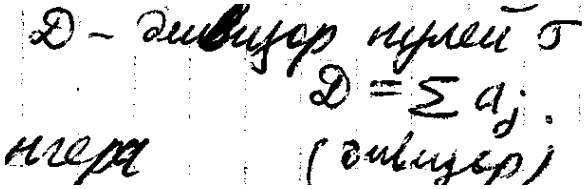
Original source details: GELFAND-P000946

Uncertain reading. The crossed-out entry is tentatively read as “Next”.

$\mathcal{D}$ — the divisor of zeros of σ

$\mathcal{D} = \sum a_j$

(divisor)



Original source details: GELFAND-P000946

Diagram and typed description. The note on the divisor in the right margin.

In fact (26) — from the Atiyah–Singer thm.

$$(\text{Coker} = 0)$$

infinite-dimensional machinery.

Magn. field here:

$$A_1 = -\partial_2\varphi, \quad A_2 = \partial_1\varphi, \quad \Delta\varphi = H. \tag{27}$$

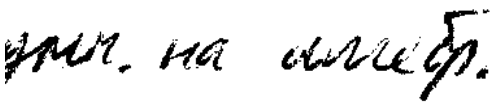
$a_1, \dots, a_N$ are determ. mod the lattice and symmetrized, i.e.

$$a \in S^N\Gamma.$$

GELFAND-P000947

The fibre (25) is a lin. space. (not immediately obvious)

All fns. Ψ_A in a fixed fibre are obt. from one by mult. by ellipt. (?) 2-periodic fns..



Uncertain reading. The abbreviation is tentatively read as the Russian “эллипт.”; “эллип.” is also possible.

Original source details: GELFAND-P000947

Corollary: Bloch fns. form, form a linear bundle.

Eigenfns. at fixed p form an N-dimensional space (!)

(the dimension is fixed)

In a uniform magn. field the assert. holds for all energy levels.

[crossed out: Question 2. ~~Where does the degeneracy come from?~~]



Original source details: GELFAND-P000947

Diagram and typed description. The whole question is crossed out.

Now let us perturb:

$$\hat{H} = \hat{H}_0 + e\,v(x,y) \qquad \text{(with the same grp. of periods).} \tag{28}$$

(29) Theorem 2. Perturb. op. — quadr. form of general type.

(2-param. family of quadr. forms in general pos. among all complex Hermitian forms).

I.e. if one excludes the set of potentials on whose Fourier coeffs. 3 eqs. are imposed, then the spectrum (Bloch eigenfns.) has the following properties

$$\hat{V}_{(p_1,p_2)}(\Psi_A) = \iint_{K_{12}} \bar{\Psi}_A\, v \Psi_A\, dx\, dy. \tag{30}$$

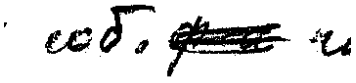
(complex Hermitian form, Ψ_A — eigenfns. of H_0)

at fixed quasimomenta.

(30) Is in general position (as an operator)

Quadr. form in general pos. — no pair of eigen [crossed out: ~~fns.~~ (?)] values coalesces

Uncertain reading. The crossed-out entry before “values” is tentatively read as “fns.”



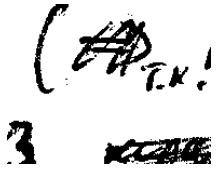
Original source details: GELFAND-P000947

The map of the torus into the set of quadr. forms is in general position.

(In the 1-dim. case — it is never in general position)

because the op. is of 2nd order.

([crossed out: ⇒ (?); inserted below: since] codim. 3, [crossed out: ~~then~~ (?)] then it does not intersect with mult. eigenvals.,



Uncertain reading. The crossed-out sign is tentatively read as ⇒.

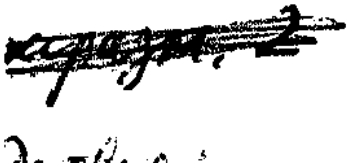
Original source details: GELFAND-P000947



Uncertain reading. The heavily crossed-out entry before “then” is tentatively read as “then”.

Original source details: GELFAND-P000947

$n = 3$ [crossed out: ~~codim. 2~~ (?)] ⇒ [crossed out: initial Cyrillic “в” (?); the rest is unresolved] it inters. at isolated points.)



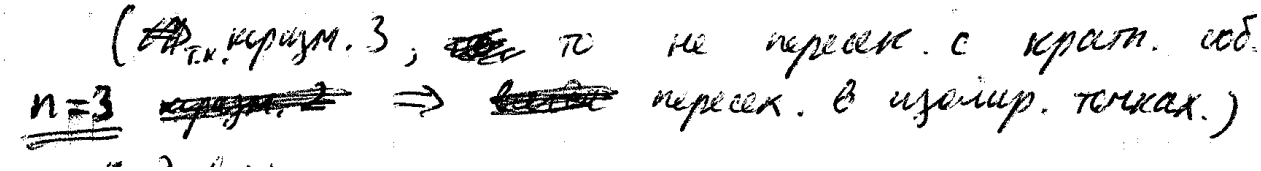
Uncertain reading. The crossed-out text is tentatively read as “codim. 2”.

Original source details: GELFAND-P000947



Undeciphered passage. After the arrow, the initial stroke resembles Cyrillic “в”. The cancellation obscures the rest.

Original source details: GELFAND-P000947



Original source details: GELFAND-P000947

Diagram and typed description. The two codimension lines, with “since” inserted below the first cancellation.

Corollary:

$n = 2$ ⇒ the bundle splits into 1-dimensional ones

I.M.G.:

The irrat. case is interesting!

(there everything is in fact noncommutative)

Idea: go into the complex domain (inside the circle!) — in a non-self-adjoint situation

i.e. carry out a regularization (parabolic)

GELFAND-P000948

[source number: 29]

Irrat. case:

$$H_{12} = \frac{\partial A_1}{\partial x_2} - \frac{\partial A_2}{\partial x_1}, \qquad A_1\,dx_1 + A_2\,dx_2.$$

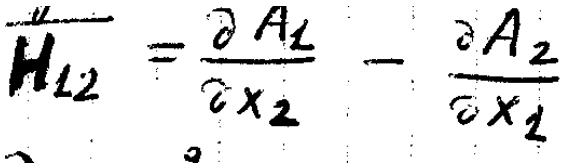


Diagram and typed description. The order of derivatives in H_{12} follows the manuscript.

Original source details: GELFAND-P000948

$$\left[\left(i\frac{\partial}{\partial x_1} - A_1\right)^2 + \left(i\frac{\partial}{\partial x_2} - A_2\right)^2\right] = \hat{H}.$$

$$H_{12}(x_1 + m_1T_1, x_2 + m_2T_2) = H_{12}(x_1, x_2)$$

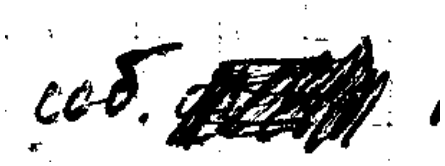
$$\Phi_{12} = \iint H_{12}\,dx_1 \wedge dx_2.$$

$$\Phi_{12} \neq 2\pi\frac{N}{M}$$

Group:

$$\hat{T}_1^*\hat{T}_2^* = \hat{T}_2^*\hat{T}_1^* \exp(-i\Phi_{12})$$

acts on the set of all fns. (fixed eigen [crossed-out entry unresolved] subspace)



Undeciphered passage. Dense hatching obscures the entry between “eigen” and “subspace”.

Original source details: GELFAND-P000948

$[H, \hat{T}_j^*] = 0$ still (choice of f_j in (12)–(13)).

At the basis of the theory [inserted above the line: of factors] of von Neumann — irreducibility ergodicity.



Diagram and typed description. A curved mark connects “of factors” above the line with the position after “theory”.

Original source details: GELFAND-P000948

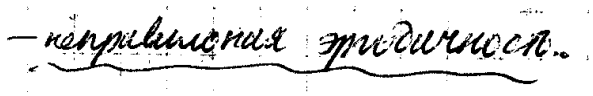


Diagram and typed description. One wavy line underlines both words.

Original source details: GELFAND-P000948

I.M.G. To know by next semester: G-structures, $\mathcal{D}$-structures and the Spencer resolution.

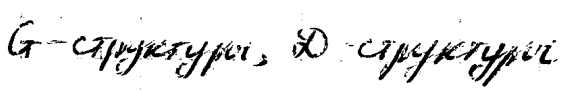


Diagram and typed description. The two structures are denoted by G and a calligraphic $\mathcal{D}$.

Original source details: GELFAND-P000948

GELFAND-P000949

[source number: 51]

9/II-81. Gelfand seminar.

MacPherson.

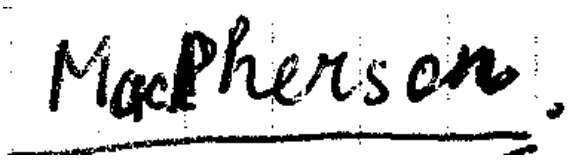


Diagram and typed description. The manuscript gives the name in Latin script.

Original source details: GELFAND-P000949

(On joint work with I. M. Gelfand.)

Problem — find interesting polyhedra in Grassmannians.

In proj. spaces polyhedra have flat faces (face $\subset$ subspace of smaller dim.)

$$P_1 \subset P_2 \subset P_3 \subset \dots \tag{0}$$

(projective)

G_n^m — n-planes in $\mathbb{R}^{m+n}$ (subspaces)

Embeddings:

$$\begin{array}{ccccccc} G_1^1 & \subset & G_1^2 & \subset & G_1^3 & \subset & \dots \\ \cap & & \cap & & \cap & & \\ G_2^1 & \subset & G_2^2 & \subset & G_1^3 & \subset & \dots \end{array} \tag{1}$$

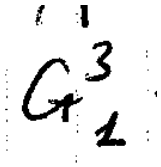


Diagram and typed description. The third entry in the lower row reads G_1^3.

Original source details: GELFAND-P000949

and so on

We want: faces — polyh. in smaller ones

Picture (0) corresp. to the upper row of (1).

Naïve method:

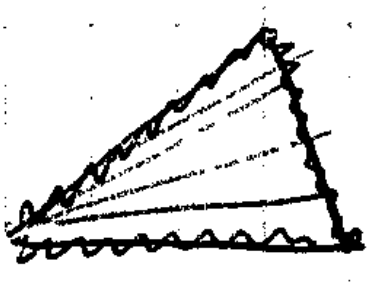
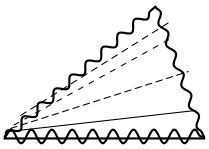


Diagram and typed description. Triangle with wavy sides and a fan of strokes. The broken interior strokes are drawn approximately.



Original source details: GELFAND-P000949

there is a triangle of 3 geod.

How to fill it in?

also with geod. (segments)

If one does this in the Grassmannian, it does not work.

Here one gets

$$\text{Rank} \geqslant 2$$



Diagram and typed description. The sign between “Rank” and 2 reads $\geqslant$.

Original source details: GELFAND-P000949

(2) Defin.

Hypersimplex Δ_n^m: set of points

$$(x_1, \ldots, x_{n+m}) \in \mathbb{R}^{n+m},$$

satisfying the rels.

$$0 \leq x_i \leq 1,$$
$$\sum_i x_i = n$$

[crossed out after n: ~~$+1$~~ (?)]



Uncertain reading. Hatching obscures the entry after n. The reading $+1$ is tentative; the letter m is also possible.

Original source details: GELFAND-P000949

(3) Def. Grassm. complex in P_{n+m-1}: choose $n+m$ points $e_1, \ldots, e_{n+m}$, not lying in a hypersubspace of smaller dim. (a hyperplane).

H^0 = conn. comp. of 1 of the subgroup of $PGL(n+m)$, which leaves these points fixed.

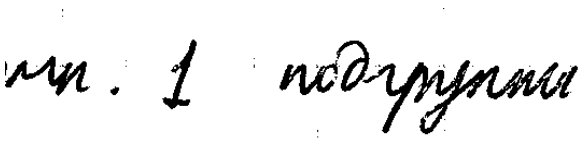


Diagram and typed description. The digit 1 follows the abbreviation for component.

Original source details: GELFAND-P000949

Grasm. complex — closure of an orbit in gen. pos. of the group H^0, acting on G_n^m. [written beneath “gen. pos.” *generic*.]

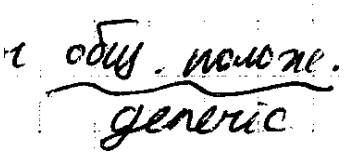


Diagram and typed description. The English gloss appears below the wavy underline.

Original source details: GELFAND-P000949

(4) *generic* — ?

this means $H^0\xi$

ξ — subspace transverse to all coord. subspaces:
if $e_1, \dots, e_{n+m}$ are taken as a basis (some points on them)
this is not the same as dim = max (!)
GELFAND-P000950

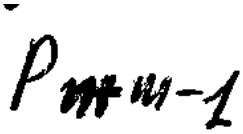
(5) Theorem. These 2 objects are diffeomorphic.
def. (3) can be given for $\forall$ semisimple group.



Original source details: GELFAND-P000950

Diagram and typed description. The abbreviated verb before “for” reads “прив.”.

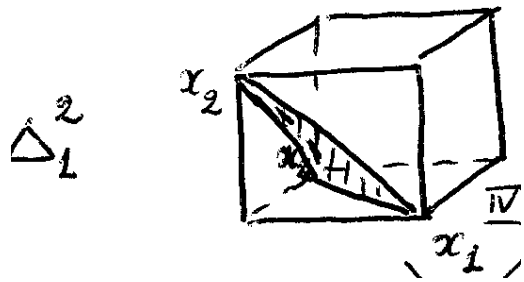
G_n^m — $(n-1)$-subspaces in P_{n+m-1} (?).



Original source details: GELFAND-P000950

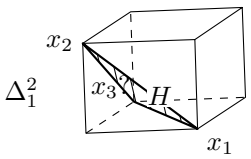
Uncertain reading. The first character of the index is heavily retraced. The reading P_{n+m-1} is tentative; P_{m+n-1} is also possible.

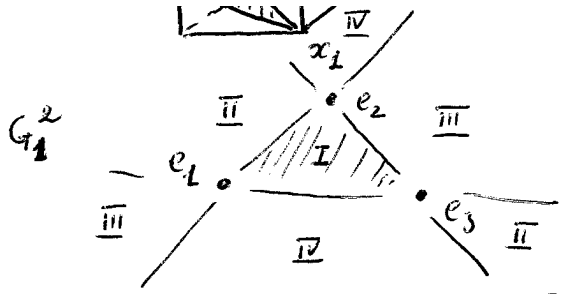
(6) Example. Δ_1^2 G_1^2



Original source details: GELFAND-P000950

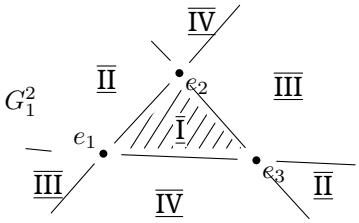
Diagram and typed description. Cube with hatched section H. The interior label, tentatively x_3, is overlapped by lines.





Original source details: GELFAND-P000950

Diagram and typed description. Three lines through the marked points e_1, e_2, e_3. The Roman numerals and hatching follow the manuscript.



transf. preserving lines as well



Original source details: GELFAND-P000950

Diagram and typed description. The character after “preserving” reads Russian “и” (“also”).

Interior of the Δ-gon — an orbit (and a conn. component of the corresp. subgrp.)
(since PGL is transit. on 4-tuples of points in general position)
Altogether — 4 triangles



Diagram and typed description. Small hatched triangle marked I.



Original source details: GELFAND-P000950

— example of a Grassm. simplex
There is a homeom., it even preserves the marks (numbers)

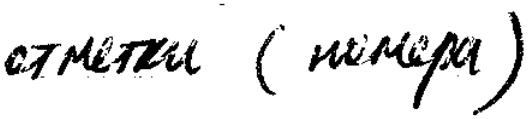


Diagram and typed description. The word before “(numbers)” reads “marks”.

Original source details: GELFAND-P000950

$\forall$ edge — a simplex in G_1^1
(7) Example. (another one !!).
G_2^1

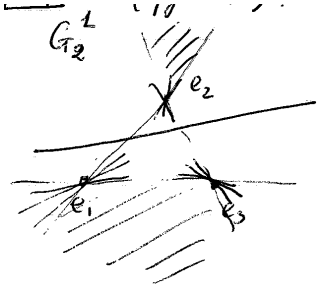
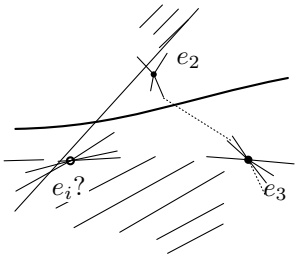


Diagram and typed description. Pencils through three points, a sloping transversal, and hatching. The faint lines are drawn approximately.



Original source details: GELFAND-P000950



Uncertain reading. The label at the lower left is tentatively e_i; the index may be 1.

Original source details: GELFAND-P000950

orbit of a line — $\forall$ lines which do not inters. the hatched region

This is a Δ-gon — with sides —

— lines passing through the vertices

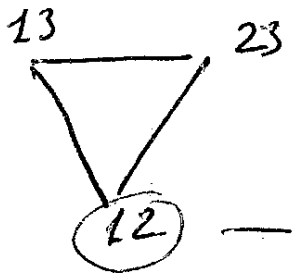
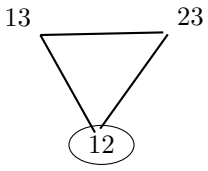


Diagram and typed description. Inverted triangle marked 13, 23, and circled 12.



Original source details: GELFAND-P000950

— corresp. to the side passing through e_1 and e_2.

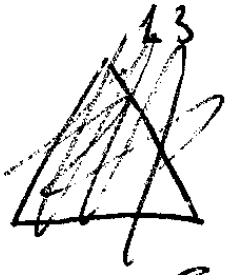
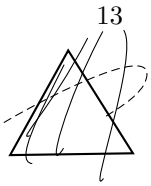


Diagram and typed description. Triangle marked 13 with curves crossing its sides. The faint curves are drawn approximately.



Original source details: GELFAND-P000950

Sides $\in G_1^1 \subset G_2^1$

Δ_2^1 — inters. of the cube with the plane $\sum x_i = 2$, with the vertices also denoted by pairs.

GELFAND-P000951

[source number: 52]

(8) Interesting example G_2^2 or Δ_2^2

[crossed out in the left margin: ~~(9)~~ (?)]



Uncertain reading. The upper crossed-out mark resembles “(9)”; “(8)” is also possible.

Original source details: GELFAND-P000951

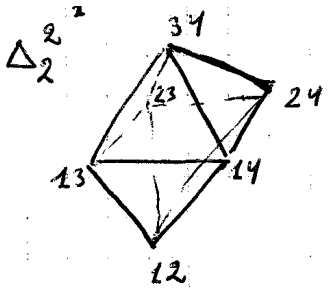
$$\sum_1^4 x_i = 2 \quad - \quad (\perp \text{ main diag. section of a 4-dim. cube})$$

[crossed out in the left margin: sign unresolved]



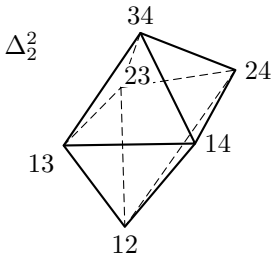
Uncertain reading. Several strokes obscure the sign to the left of the figure.

Original source details: GELFAND-P000951



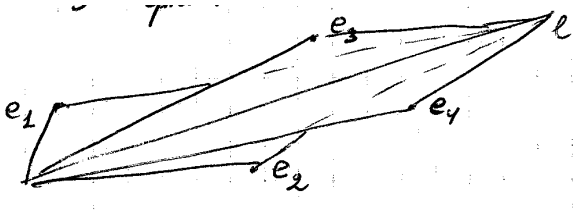
Original source details: GELFAND-P000951

Diagram and typed description. Labelled sketch of Δ_2^2. The faint interior connections are drawn approximately.



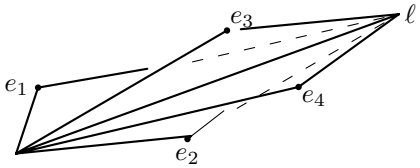
G_2^2 – 4-dimensional

H^0 – 3-dimensional.



Original source details: GELFAND-P000951

Diagram and typed description. Sketch with points e_1, e_2, e_3, e_4 and label ℓ. The faint broken strokes are drawn approximately.



$K(\ell) =$ cross-ratio of 4 planes

(anharm. ratio) passing through ℓ and the sel. points

(take the quotient of everything by $\ell \Rightarrow$ get it in the plane)

$$H^0\text{-orbit} \subset K(\ell) = \text{const}$$

$$\begin{array}{c|c|c|c|c} G_2^2 & \multicolumn{3}{c|}{\left\{\begin{array}{ccc|c} 12 & 13 & 14 & 1 \\ 21 & 23 & 24 & 2 \\ 31 & 32 & 34 & 3 \\ 41 & 42 & 43 & 4 \end{array}\right.} \\ \cup & & & \\ G_2^2 \supset & & & \end{array}$$

$$\begin{array}{c|c|c|c|c} G_2^1 \supset & \multicolumn{3}{c|}{\left\{\begin{array}{ccc|} 14 & 42 & 21 & \\ 13 & 32 & 21 & \\ 14 & 43 & 31 & \\ 24 & 43 & 32 & \end{array}\right.} \\ \cap & & & \\ G_2^2 & & & \end{array}$$

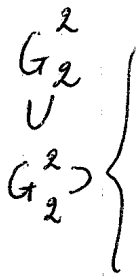


Diagram and typed description. Both upper labels read G_2^2, with a U-shaped sign between them.

Original source details: GELFAND-P000951

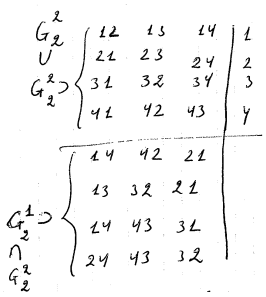
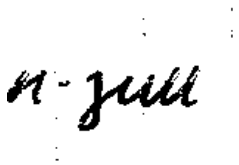


Diagram and typed description. The two blocks of the manuscript table, with the order of the numbers preserved.

Original source details: GELFAND-P000951

Each one — in its own [abbreviation unresolved] $K(\ell) = \text{const}$



Uncertain reading. The abbreviation after “its own” is tentatively read as “мн-зии”. Its exact expansion is unresolved.

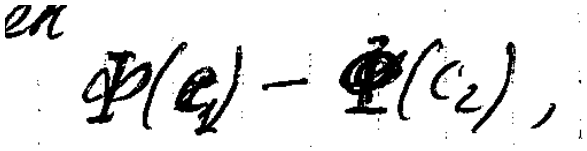
Original source details: GELFAND-P000951

(9) Theorem. The volume of the set, encl., between 2 Grassm. simplices

$$K(\ell) = c_1, \quad K(\ell) = c_2$$

equals

$$\Phi(c_1) - \Phi(c_2),$$



Uncertain reading. The second Φ is retraced. Its earlier form is unresolved.

Original source details: GELFAND-P000951

where

$$\Phi(x) = L_1(x) - L_1(1-x), \quad L_1(x) = \sum_{n=1}^{\infty} \frac{x^n}{n^2}, \quad 0 < c_1 < c_2 < 1$$

(integr. logarithm)

GELFAND-P000952

(10) Theorem. Generally speaking, [inserted above the line: combinatorial] homeomorphism

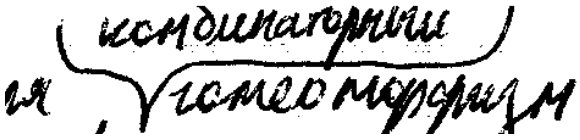


Diagram and typed description. “Combinatorial” is above the line, linked by a curved mark to the start of “homeomorphism”.

Original source details: GELFAND-P000952

$\Delta_n^m \hookrightarrow$ Grassm. complex G_n^m

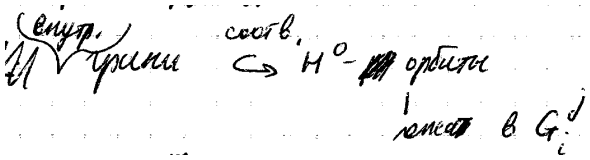
preserves a number of further structures.

(11) [crossed out: ~~Theorem~~]



Original source details: GELFAND-P000952

Diagram and typed description. The underlined word “Theorem” remains legible beneath the hatching.



Original source details: GELFAND-P000952

Diagram and typed description. Correspondence of faces and H^0-orbits. The stroke under “orbits” has no visible arrowhead.

inter. corresp.

Faces $\hookrightarrow H^0$-[?] orbits

lie in G_i^j

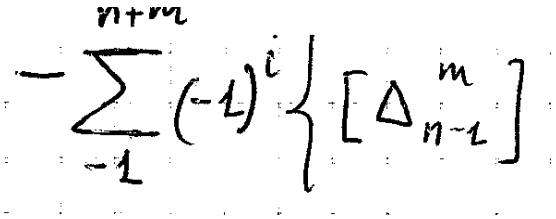


Original source details: GELFAND-P000952

Uncertain reading. The crossed-out entry after H^0 and the hyphen is unresolved.

(12)

$$\partial[\Delta_n^m] = \sum_{i=1}^{n+m} (-1)^i \{[\Delta_n^{m-1}] \mid i\text{-th coord.} = 0\} - \sum_{-1}^{n+m} (-1)^i \{[\Delta_{n-1}^m]$$

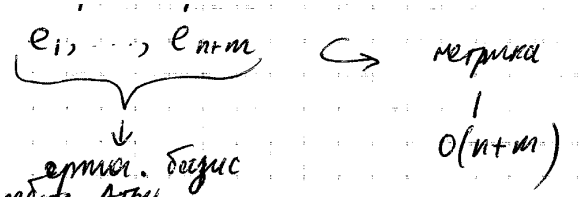


Original source details: GELFAND-P000952

Undeciphered passage. The lower limit in the second sum reads -1. The entry breaks off after $[\Delta_{n-1}^m]$.

(13) Theorem. $\forall$ harm. form in G_n^m vanishes on any Grassm. simplex.

The metric is ass. to be compatible in some way with the coordinates.



Original source details: GELFAND-P000952

Diagram and typed description. A brace under the list leads to the basis. The stroke below “metric” has no visible arrowhead.

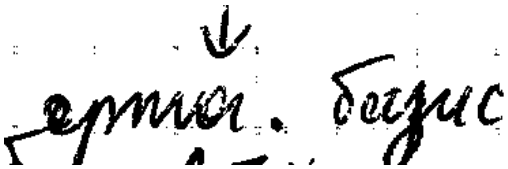
$\underbrace{e_i, \ldots, e_{n+m}}_{\downarrow}$ $\longrightarrow$ metric

orthog. basis (?) $O(n+m)$



Original source details: GELFAND-P000952

Diagram and typed description. The first e has a dotted subscript i.



Original source details: GELFAND-P000952

Uncertain reading. The abbreviation is tentatively “orthog.”; “orthon.” is also possible.

On Atiyah’s work.

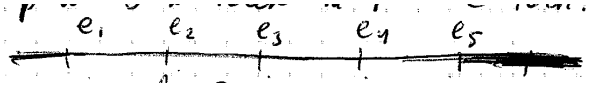


Original source details: GELFAND-P000952

Diagram and typed description. The manuscript spells the name “Атьи”.

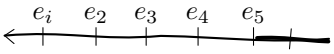
Hence the funct. eq. for the dilogarithm.

Space of 5-tuples of points on P^1 up to proj. transf.



Original source details: GELFAND-P000952

Diagram and typed description. Line with five labels and an unlabelled tick on the right. The darkened end is drawn approximately.



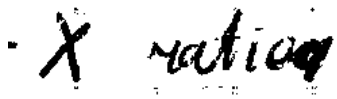


Original source details: GELFAND-P000952

Diagram and typed description. The first e is labelled with a dotted i.

This space is 2-dimensional.

$$k_1 = \text{X ratio } (e_2\, e_3\, e_4\, e_5)$$
$$k_2 = \text{— ” — } (e_1\, e_3\, e_4\, e_5)$$



Original source details: GELFAND-P000952

Uncertain reading. The sign before “ratio” is tentatively read as X. The shapes of X and $\times$ are close.

. . .

$$\varphi(x) = \text{Li}(x) - \text{Li}(1-x), \qquad \text{Li}(x) = \sum_1^\infty \frac{x^n}{n^2}.$$

$$0 = d\varphi(k_1) - d\varphi(k_2) + d\varphi(k_3) - d\varphi(k_4) + d\varphi(k_5) \tag{14}$$

(Funct. eq. for the dilogarithm)

This follows from the previously formul. theorems.

G_3^2 — 3-dim. planes in $\mathbb{R}^5$.

Consider $P_1(\xi)$ (1st Pontryagin class of the tautol. 2- or 3-dim. bundle on the Grassm. G_3^2).

There is exactly 1 harm. 4-form.

GELFAND-P000953

[source number: 53]

$$\underset{\|}{d}\pi_* P_1(\xi) = \pi_* \underset{\|}{d} P_1(\xi) + \sum \pi_{i*}^{\partial} P_1(\xi)$$
$$0 \qquad\qquad 0$$

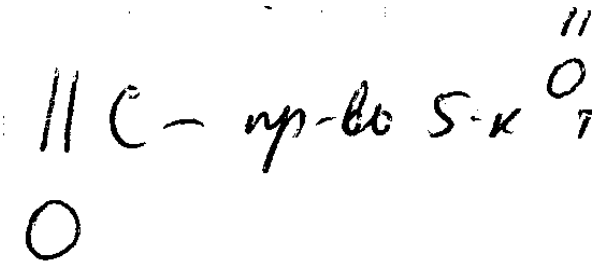


Diagram and typed description. Two strokes and a zero appear below each of the two d symbols.

Original source details: GELFAND-P000953

C — space of 5-tuples of points

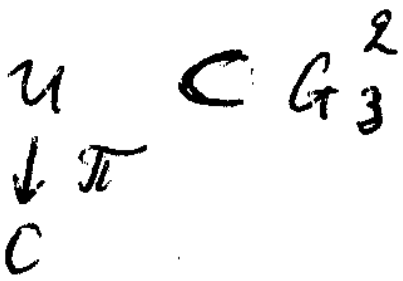


Diagram and typed description. The sign to the left of $\subset$ resembles lowercase u; uppercase U is also possible.

$$\begin{array}{ll} u? & \subset G_3^2 \\ \downarrow \pi & \\ C & \end{array}$$

Original source details: GELFAND-P000953

The last term is calc. with the aid of combinatorics

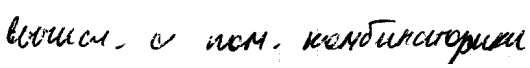


Diagram and typed description. The words before “combinatorics” read “with the aid of”.

Original source details: GELFAND-P000953

Generalization.

Let $(\mathbb{C}^*)^n$ act algebraically on a homogeneous algebr. [crossed out: entry unresolved] space with a fixed point



Undeciphered passage. The entry between “algebr.” and “space” is heavily crossed out.

Original source details: GELFAND-P000953

Then

1) $\forall$ orbit of $\left((\mathbb{R}^+)^*\right)^n$ is homeom. to a polyhedron. (theorem on toric embeddings)

(Diffeom. on $\forall$ orbit, but at the vertices it may have ∞ curvature; on the whole space — only a homeom.)

2) $\forall$ harm. form [crossed out: entry unresolved] invar. under $T^n \subset (\mathbb{C}^*)^n$



Undeciphered passage. Several strokes obscure the entry after “form”.

Original source details: GELFAND-P000953

[inserted below “harm. form”: (with respect to $\forall$ Kähler metric)]



Diagram and typed description. The condition on the metric is written on the line below.

Original source details: GELFAND-P000953

is equal to 0 on $\forall$ $\left((\mathbb{R}^+)^*\right)^n$-orbit.

(15) Theorem. (Atiyah).

M — symplect. manif., $f_1, \ldots, f_n \colon M \longrightarrow \mathbb{R}$ functions.



Diagram and typed description. The Russian abbreviation after “symplect.” reads “мн-зие” (manifold).

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$$\{f_i, f_j\} = 0$$

df_i all vanish to 0 at some point.

We get a map

$$M \underset{f_1,\ldots,f_n}{\longrightarrow} \mathbb{R}^n$$

Here

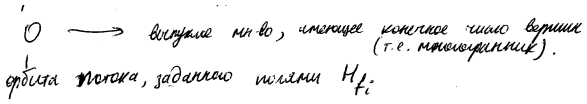


Diagram and typed description. An arrow leads from O to the convex set. A stroke below O leads to the orbit label.

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O ⟶ convex set having finitely many vertices (i.e. a polyhedron).

|

orbit of the flow given by the fields H_{f_i}



Diagram and typed description. Two strokes with a dot between them.

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Problem: relation to integr. by quadratures.